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Review article

Incorporating machine learning and statistical methods to address maternal healthcare disparities in US: A systematic review

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ABSTRACT

Background: Maternal health disparities are recognized as a significant public health challenge, with pronounced disparities evident across racial, socioeconomic, and geographic dimensions. Although healthcare technologies have advanced, these disparities remain primarily unaddressed, indicating that enhanced analytical approaches are needed.

Objectives: This review aims to evaluate the impact of machine learning (ML) and statistical methods on identifying and addressing maternal health disparities and to outline future research directions for enhancing these methodologies.

Methods: Following the PRISMA guidelines, the review of studies employing ML and statistical methods to analyze maternal health disparities within the United States was conducted. Publications between January 1, 2012, and February 2024 were systematically searched through PubMed, Web of Science, and ScienceDirect. Inclusion criteria targeted studies conducted within the U.S., peer-reviewed articles published during the period, research covering the postpartum period up to one year post-delivery, and studies incorporating both maternal and infant health data with a focus primarily on maternal outcomes.

Results: A total of 147 studies met the inclusion criteria for this analysis. Among these, 129 (88 %) utilized statistical methods in health sciences to analyze correlations, treatment effects, and public health initiatives, thus providing vital, actionable insights for policy and clinical decisions. Meanwhile, 18 articles (12 %) applied ML techniques to explore complex, nonlinear relationships in data. The findings indicate that while ML and statistical methods offer valuable insights into the factors contributing to health disparities, there are limitations regarding dataset diversity and methodological precision. Most studies concentrate on racial and socioeconomic inequalities, with fewer addressing the geographical aspects of maternal health. This review emphasizes the necessity for broader dataset utilization and methodology improvements to enhance the findings' predictive accuracy and applicability.

Conclusions: ML and statistical methods show great potential to transform maternal healthcare by identifying and addressing disparities. Future research should focus on broadening dataset diversity, improving methodological precision, and enhancing interdisciplinary efforts.

1. Introduction

Women who are pregnant or have recently given birth are at a much higher risk of experiencing serious health issues, including premature death [1]. According to the World Health Organization (WHO), 99 % of maternal deaths occur in low- and middle-income countries due to barriers such as poor healthcare infrastructure and socioeconomic inequalities [2]. For instance, despite advancements in modern medicine, the unequal distribution of healthcare resources globally continues to delay progress in achieving equitable maternal health outcomes [2].

In the United States, adverse health outcomes occur at higher rates than in any other developed nation, and these disparities continue to grow [3]. Maternal mortality in the U.S. has more than doubled over the past three decades, with some demographic groups experiencing

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1386-5056/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

mortality rates three times higher than others, regardless of income or education level [4,5]. These maternal health disparities reflect variations in pregnancy outcomes across racial and ethnic groups, often due to systemic biases and unequal access to healthcare [6]. Contributing factors include disparities in access to high-quality care, implicit bias in medical treatment, and broader structural issues like racism and socioeconomic inequality [5].

Addressing these disparities requires an integrated approach that includes healthcare system improvements, community engagement, and targeted research [4]. Innovations are critical in this initiative, particularly through machine learning (ML) and advanced statistical analysis. By analyzing large datasets, ML tools can identify patterns, predict outcomes more precisely, and help address health inequities. For instance, ML algorithms improve disease detection and data accuracy by automating diagnostics, reducing human error [7]. Moreover, ML methods such as Convolutional Neural Networks (CNNs) have revolutionized medical imaging, enabling precise detection of conditions like skin cancer [8,9].

Table 1 summarizes previous systematic reviews that have explored various aspects of maternal health disparities. It captures various studies

Table 1
Summary of Previous Systematic Reviews on Maternal Health Disparities.

Systematic Review	Review Period	Number of Reviewed Articles	Geographical Location	Focus	Disparity Area	Clinical Implication
Humphries et al. [10]	Up to July 22, 2015	24	USA, UK, and Australia	Explored how race and ethnicity affect IVF success rates and outcomes.	IVF success rates vary by race, with White women generally having higher success compared with Black, Hispanic, and Asian women.	highlights the need for personalized IVF strategies and further research to address disparities in outcomes across racial and ethnic groups.
Ashman et al. [11]	Up until October 2015	27	Australia, USA, and Canada	Studying nutritional interventions for pregnant Indigenous women in OECD countries to identify success factors.	Examines health and socioeconomic disparities between Indigenous and non-Indigenous groups, emphasizing prenatal and infant nutrition.	Emphasizes the need for culturally tailored, community-involved, and sustainable nutrition interventions to improve health outcomes for pregnant Indigenous women.
Vats et al. [1]	March 1, 2020, to July 11, 2020	86	Asia, Europe, North America, Oceania, and South America.	Examined how pre-pregnancy BMI influences health risks for mothers and their newborns	Highlights how BMI thresholds, geographic location, BMI data sources, sample size, and data disparities influence maternal and fetal/neonatal health outcomes	Elevated maternal BMI increases the risks of adverse outcomes, highlighting the need for weight management and tailored prenatal care.
Shrivastava et al. [12]	January 2011 to May 2022	11	India	Concentrated on pinpointing obstacles and opportunities in obtaining maternal healthcare during pregnancy in India's urban slums	The article emphasized the unequal use of maternal healthcare services in urban slums, shaped by socio-economic and demographic influences	Improved education and AI tools are needed to overcome access, cost, and literacy barriers in urban slum maternal healthcare.
Bizuayehu et al. [13]	Last updated on April 15, 2022, with initial publication on May 28, 2018	10	Australia	Examined how living in urban versus rural areas in Australia affects premature births, low birth weight, and cesarean section rates.	The research highlighted differences in pregnancy outcomes and associated risk factors between rural/remote and urban regions in Australia.	Improved healthcare access and antenatal care in rural Australia are vital to reducing disparities in preterm births and low birth weight.
Biset et al. [14]	2000 to 2021	35	Ethiopia	Examining colostrum avoidance and its factors among Ethiopian mothers.	Examines regional and socio-economic disparities in colostrum avoidance in Ethiopia.	highlights the need for improved maternal counseling and education on colostrum's benefits to reduce colostrum avoidance among Ethiopian breastfeeding mothers.
Guo et al. [15]	Up to November 2022	45	Global	Explores the impact of prenatal BPA exposure on neonatal health outcomes.	Explores disparities in neonatal health outcomes due to prenatal BPA exposure.	highlights the need to minimize maternal BPA exposure during pregnancy to reduce risks of adverse neonatal outcomes, emphasizing further research to confirm and clarify these associations.
Stanhope et al. [16]	Up until February 13, 2023	19	USA, Canada, Australia, Ireland, Japan, Kenya, Netherlands, South Africa, and Sweden	Examines the impact of social risk screening in OBGYN care on outcomes.	Explores disparities in SDOH screening during prenatal care, focusing on intimate partner violence.	Emphasizes the need for structured, supported SDOH screening protocols in OBGYN care, ensuring effective referrals and addressing social risk factors to improve patient outcomes.
Ekezie et al. [17]	Up to June 2023	45		Examines maternal and perinatal health outcomes in GRT communities, focusing on key risk factors.	Highlights perinatal disparities, with GRT women facing higher preterm births and infant mortality.	Address inequities with tailored care and better access to improve GRT maternal and perinatal outcomes.
Bjerkhaug et al. [18]	January 1, 1996, to July 31, 2023	26	13 European countries	Assesses the immunogenicity and safety of maternal GBS vaccines in clinical trials.	Explores disparities in vaccine efficacy and safety among pregnant women.	highlights that maternal GBS vaccines show good immunogenicity and safety, emphasizing the potential to prevent neonatal GBS disease and the importance of further trials for broader validation.

examining factors such as race, ethnicity, nutrition, and geographic location concerning maternal health outcomes and disparities. These summaries help clarify the breadth of past investigations and the continuing need for a detailed analysis as proposed in this review.

By synthesizing insights from these reviews, this systematic review aims to provide an overview of the state of ML and statistical methods applied to maternal health within the US, which can guide future improvements in maternal healthcare services and delivery.

The main objectives of this review are:

- Examining the usage of ML and statistical methods, including predictive modeling, data analytics, and decision-making to improve maternal health outcomes and predict complications.
- Exploring how socioeconomic and socio-demographic disparities within the US impact the quality of maternal healthcare services and outcomes.
- Identifying new research opportunities and unexplored areas where statistical analysis and ML could address disparities in maternal health.

2. Materials and methods

This research followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines established by Moher et al. [19]. Additionally, it was registered with the International Prospective Register of Systematic Reviews (PROSPERO) under the National Institute for Health Research, with the registration number CRD42024578664.

2.1. Eligibility criteria

The inclusion of studies was determined according to criteria jointly defined and agreed upon by HA and AR.

Inclusion Criteria:

- Studies conducted within the United States.
- Published articles (peer-reviewed) are available as of February 29, 2024.
- Articles analyzing the postpartum period up to and including one year post-delivery.
- Research that includes both maternal and infant health data, with a primary focus on the mother's health outcomes
- Studies published between January 1, 2012, and February 2024.

Exclusion Criteria:

- Studies conducted outside the United States.
- Working papers, along with articles under review or unpublished as of February 29, 2024.
- Articles analyzing the postpartum period extending beyond one year post-delivery.
- Research focusing exclusively on infant health without integrating maternal health data.
- Articles that do not directly address disparities.

2.2. Information sources

The study conducted a search of three electronic databases: PubMed, Web of Science, and ScienceDirect. The search process was focused on research published from January 1, 2012, to February 2024. This timeframe highlights the increasing interest in advanced analytical techniques to enhance maternal healthcare outcomes.

2.3. Study Selection

A search strategy was developed after defining the research question variables and establishing selection criteria for the articles in the databases. This strategy utilized key terms and logical operators (AND, OR) for PubMed, ScienceDirect, and Web of Science databases. The key terms included: "Maternal," "Pregnancy," "Disparity," "Inequality," "Inequity," "Artificial Intelligence," "AI," "Machine Learning," and "ML.".

The following search query was applied to each of the three databases:

(((Maternal OR Pregnancy) AND (Disparity OR Inequality OR Inequity)) AND (Artificial Intelligence OR AI OR Machine Learning OR ML)).

2.4. Screening and Selection process

The Rayyan QCRI platform was used for screening articles [20]. After duplicates were removed, two trained reviewers (HA, AR), both experienced in literature analysis, independently evaluated the titles and abstracts. Any disagreements were resolved through discussion or consulting a third reviewer (SCM). Full-text reviews of all potentially eligible articles were then conducted according to the inclusion and exclusion criteria.

2.5. Data collection process

The extracted data included the key participant characteristics, study objectives, research design, sample size, and analytical tools were also documented. Additionally, metrics on relevant variables, model parameters, and outcomes were collected, with all findings compiled in a shared sheet document for collaborative review and analysis.

2.6. Assessment of risk of bias

This systematic review applied the revised Cochrane risk-of-bias tool (RoB 2) and the Risk of Bias in Non-Randomized Studies (ROBINS-I) to assess biases in randomized and non-randomized studies, respectively. These tools evaluated biases across multiple domains, with judgments ranging from low to critical risk. Two independent reviewers (HA, AR) performed the evaluations, resolving any disagreements through discussion.

3. Results

3.1. Study selection

The study searched databases including PubMed, Web of Science, and ScienceDirect, initially finding 4,581 articles on maternal health disparities and technological interventions. Following PRISMA guidelines (illustrated in Fig. 1), duplicates were removed, and articles were screened based on ML and statistical methods relevance. Ultimately, 147 articles were selected for a detailed review.

3.2. Risk of bias of included studies

In evaluations conducted with the RoB 2 tool, as depicted in Fig. 2, randomized controlled trials generally demonstrate a low risk of bias across various domains. In contrast, applying the ROBINS-I tool to non-randomized studies reveals that 88 % of studies are classified as moderate risk.

3.3. Study characteristics

3.3.1. Cohorts analyses

3.3.1.1. Pregnant Women. The 'Pregnant Women' cohort analysis is supported by substantial research, including the study by Nguyen et al. [21], which encompasses an extensive sample size of over 56 million participants. This cohort is well-supported by a wide range of research,

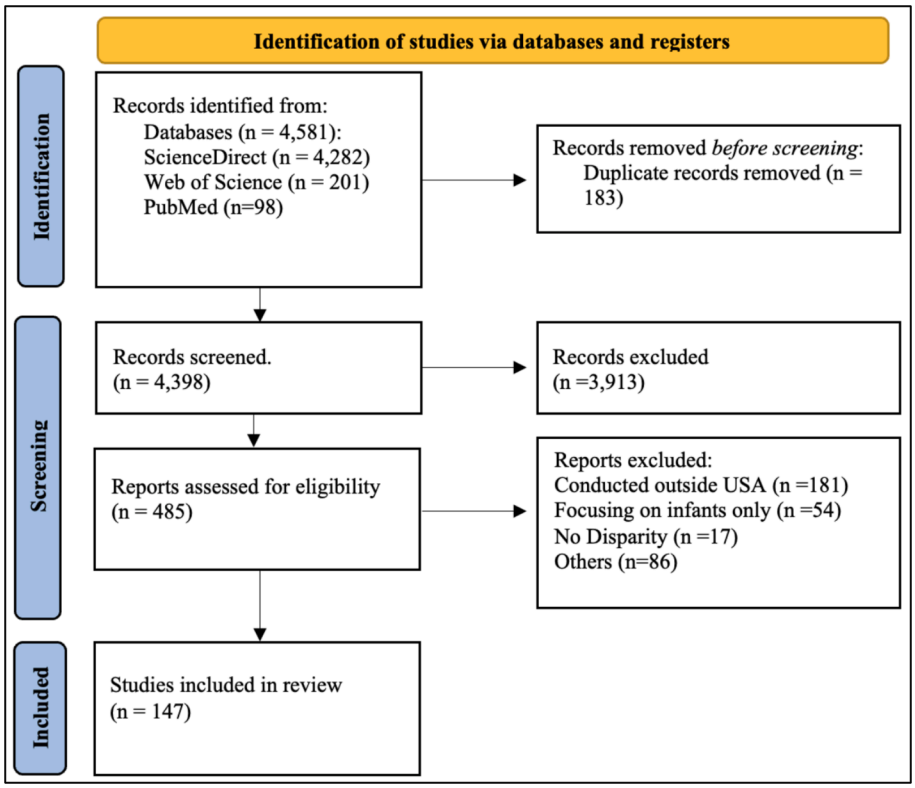

Fig. 1. Literature Search and Selection Process Flowchart according to PRISMA guidelines.

as exemplified in Unal et al. [22], covering biological and clinical measures alongside demographic and socioeconomic factors, maternal health and lifestyle, and pregnancy-related factors, among others. Studies such as those offer a panoramic view of the factors influencing maternal health, from psychosocial and behavioral measures highlighted in Giurgescu et al. [23] to border environmental and community factors investigated in Zahran et al. [24].

3.3.1.2. Postpartum Women. The 'Postpartum' cohort is analyzed, covering research from the immediate to extended postpartum periods. This includes studies such as Goyal et al. [25], which analyzes a large sample of 7,807 women up to 12 months postpartum, and the research by Guan et al. [26], focusing on the first day after delivery with 365 new mothers. Furthermore, Park et al. [27] conducted a comprehensive follow-up of over half a million participants for 60 days, while Martin et al. [28] focused on a smaller group of 90 individuals over a 42-day period.

3.3.1.3. Pregnant Women and Infants. Several studies have examined maternal and infant health concerning environmental and social factors. Hoepner et al. [29] investigated bisphenol A (BPA) levels, while Bloom et al. [30] explored the effects of maternal phthalate exposure on birth outcomes.

Kanichy et al. [31] focused on breastfeeding disparities in North Dakota, Sarullo et al. [32] and Hernandez-Castro et al.[33] assessed the impacts of socioeconomic status and prenatal exposure to organophosphate esters on fetal development and birth weight. Additionally, Jara et al. [34] analyzed how pregnancy hormones and maternal and infant birth weights interact, considering racial and BMI influences. Jacobson et al. [35] studied the relationship between fetal growth patterns and maternal health and ethnicity.

3.3.1.4. Postpartum and Infants. The 'Postpartum and Infants' group is distinctly captured through a key study conducted by Chavez et al. [36], exploring the link between social needs and postpartum depression (PPD). This study, with a cohort of 3,616 participants from 2021 through 2022, focuses on the prevalence of PPD symptoms during routine infant well-child visits. It investigated the connection between mother's reported social needs and the manifestation of PPD symptoms, offering valuable perspectives on the socio-psychological factors affecting the mental health of new mothers.

Table 2 outlines the study cohorts, detailing their sample, with Demographic and Socioeconomic factors (129 studies) and Maternal Health and Lifestyle factors (107 studies) being the most prevalent.

3.4. Disparity in maternal health

3.4.1. Disparity areas

Maternal health disparities are influenced by a variety of factors, which can be organized into several key categories:

- **Racial/Ethnic and Socioeconomic Disparities:** This category comprises research investigating disparities in health outcomes, exposures, or healthcare access depending on race, ethnicity, or socioeconomic status.
- **Disparities in Birth Outcomes and Pregnancy Complications:** These studies look at childbirth and pregnancy outcome gaps, including preterm birth rates, eclampsia, cesarean deliveries, and differences among diverse populations.
- **Disparities Related to Chemical and Environmental Exposures:** These studies explore how chemical and environmental exposures impact pregnant women's health, considering substances like BPA, phthalates, and pollutants across diverse populations.
- **Disparities in Healthcare Access and Treatment:** This category looks at healthcare disparities among groups, including differences in prenatal care, treatment options, and experiences influenced by factors like race and insurance coverage.
- **Disparities in Maternal and Neonatal Health:** This category explores health outcome disparities for both mothers and babies,

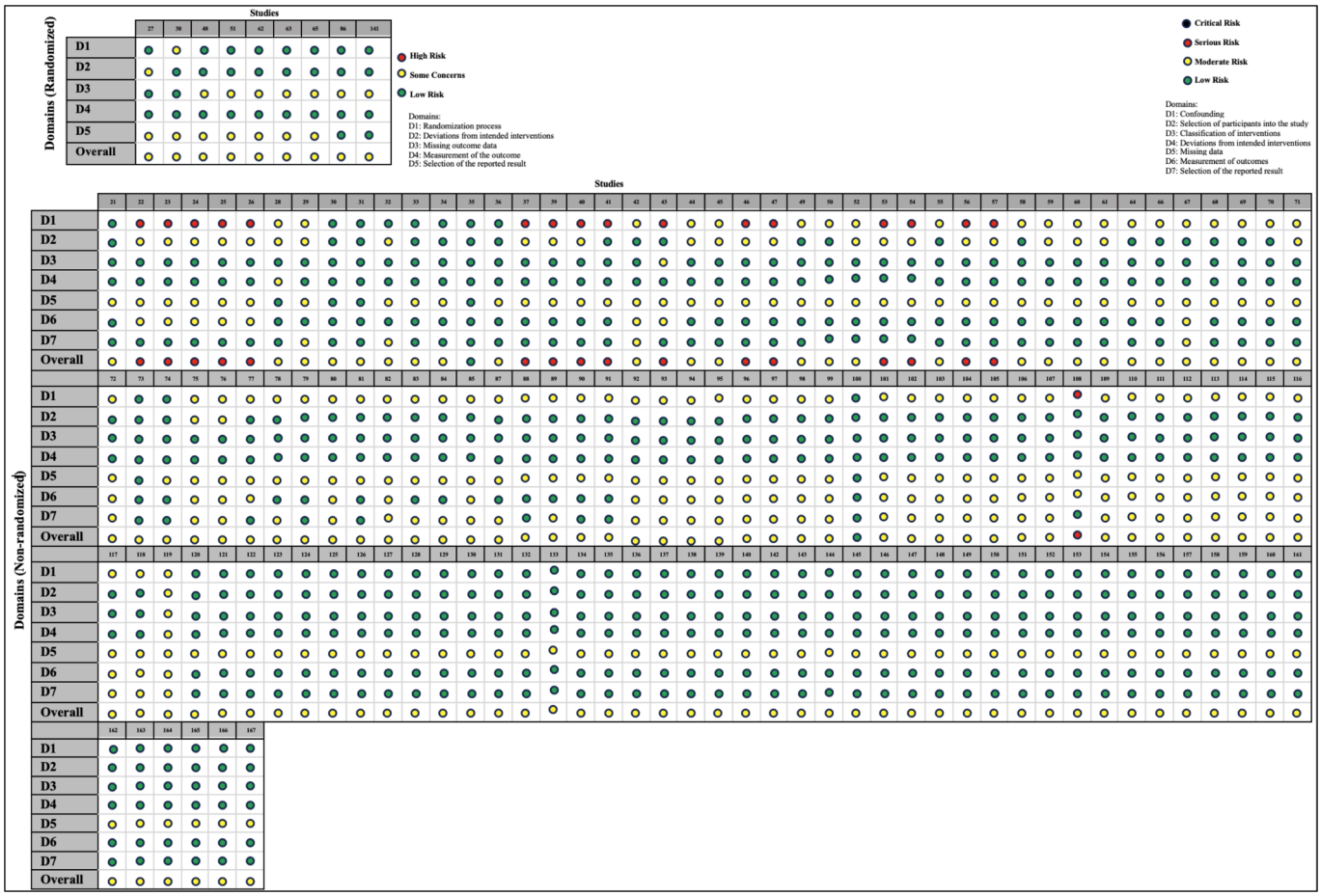

Fig. 2. Comparative Risk of Bias Assessment in Randomized and Non-Randomized Studies Using RoB 2 and ROBINS-I Tools.

including conditions like gestational diabetes and hypertension and neonatal factors such as birth weight and infant mortality

- **Disparities in Behavioral and Lifestyle Factors:** These studies investigate how lifestyle choices during pregnancy, like diet, exercise, smoking, and alcohol consumption, affect maternal health across diverse backgrounds
- **Disparities in Mental Health:** This category studies mental health changes during and after pregnancy, focusing on issues like depression and anxiety across different demographics

Table 3 presents studies categorized by their respective domains. Notably, most studies primarily focus on racial/ethnic and socioeconomic disparities, accounting for 39 %.

3.4.2. *Disparity metrics*

Disparity metrics are essential for quantifying and understanding maternal health disparities. These tools enable researchers and policymakers to pinpoint areas with significant disparities and evaluate the effectiveness of interventions designed to mitigate these gaps.

The Odds Ratio (OR) is the most used metric, indicating its significance in assessing the likelihood of outcomes across various groups. Confidence intervals (CI) and adjusted odds ratios (AOR) are also commonly used, emphasizing their role in improving disparity estimates. Other measures, such as beta coefficients and risk ratios, provide further detail, although they are applied less frequently.

3.5. *Machine learning and statistical analysis implementation*

This section reviews the methodological approaches in maternal health disparities research, focusing on statistical analyses and ML techniques. Of the 147 articles examined, 129 (88 %) used statistical methods in health sciences to analyze correlations, treatment effects, and public health initiatives, providing vital, actionable insights for policy and clinical decisions. Meanwhile, 18 articles (12 %) applied ML techniques, reflecting an increasing adoption to explore complex, nonlinear data relationships.

3.5.1. *Statistical analysis studies*

Fig. 3 illustrates the statistical methods employed to study maternal health disparities. Regression techniques, including Logistic Regression and Multivariable Linear Regression, are commonly utilized, underscoring their significance in exploring the connections between different variables and maternal health results. Logistic regression, for instance, is widely applied in studies like those by Kitsantas et al. [37], demonstrating its effectiveness in analyzing binary outcome data. Additionally, Linear Regression and Cox Proportional Hazards models are used to examine continuous outcomes and time-to-event data, respectively.

Sophisticated models such as Generalized Estimating Equations and Structural Equation Modeling, as used by Choi et al. and Yu et al. [73,99], capture the multi-level and longitudinal patterns of maternal health data. Using Bioinformatics Tools and Receiver Operating Characteristic (ROC) Curves, particularly in predictive analytics, indicate a growing trend in data science and maternal health interface. Measures of correlation and association, including Spearman's and Pearson's, are essential for identifying the strength and direction of relationships between variables, playing a pivotal role in exploratory research into disparities in maternal health. This application effectively applies advanced statistical techniques and predictive analytics to understand

Table 2
Summary of Study Cohorts: Sample Sizes, Duration, and Key Characteristics.

Publications	Sample Size	Demographic and Socioeconomic	Maternal Health and Lifestyle	Pregnancy-Related	Environmental and Community	Biological and Clinical Measures	Health Care and Policy	Psychosocial and Behavioral Measures	Infant and Child Health Factors
Kitsantas et al. [37](A)	5000	✓	✓	✓					
Unal et al. [22](A)	27	✓	✓	✓		✓			
Dunlop et al. [38](A)	160	✓	✓				✓		
Christian et al. [39](A)	56	✓	✓	✓		✓			
Wegienka et al. [40](A)	777	✓			✓				✓
Goyal et al. [25](B)	7,807	✓	✓	✓					
Wallace et al. [41](A)	866	✓		✓	✓	✓			
Dolbier et al. [42](B)	206	✓	✓	✓					
Hoepner et al. [29](C)	568	✓	✓			✓			
Creanga et al. [43](A)	3,476,392	✓	✓				✓		
Menon et al. [44](A)	191	✓		✓		✓			
Zahran et al. [24](A)	75,501	✓	✓	✓	✓				
Drassinower et al. [45](A)	1,009	✓	✓	✓					✓
Chang et al. [46](A)	291	✓				✓			
McQueen et al. [47](A)	4,045	✓	✓						
Stephenson et al. [48](A)	1,308	✓		✓					
Bruckner et al. [49](A)	1,426,488	✓	✓						
Gillespie et al. [50](A)	76	✓	✓	✓					
Hansen et al. [51](A)	900		✓						
Brown et al. [52](A)	1,463	✓					✓		
Zamora-Kapoor et al. [53](A)	71,075	✓	✓				✓		
Lemon et al. [54](A)	1,055,359	✓	✓						✓
Giurgescu et al. [23](A)	96	✓	✓			✓		✓	
Hill et al. [55](A)	3,289	✓	✓			✓			
Hsu et al. [56](A)	62,588	✓				✓	✓		
Miller et al. [57](A)	119	✓	✓	✓			✓		
Gillespie et al. [58](A)	89	✓	✓	✓				✓	
Most et al. [59](A)	66	✓	✓						✓
Specker et al. [60](A)	10,299	✓	✓						
Weber et al. [61](A)	336,214	✓	✓		✓				
Patten et al. [62](A)	352	✓	✓						
Agrawala et al. [63](A)	1,228	✓	✓	✓					
Koning and Ehrenthal [64](A)	131,418	✓			✓				✓
Yusuf et al. [65](A)	345	✓	✓			✓			
Johnson et al. [66](A)	1,701	✓	✓	✓					

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Table 2 (*continued*)

Publications	Sample Size	Demographic and Socioeconomic	Maternal Health and Lifestyle	Pregnancy-Related	Environmental and Community	Biological and Clinical Measures	Health Care and Policy	Psychosocial and Behavioral Measures	Infant and Child Health Factors
Siegel et al. [67](B)	2,948	✓	✓						
Bloom et al. [30](C)	310	✓	✓	✓		✓			
Gerson et al. [68](A)	472	✓	✓	✓					
Carroll et al. [69](A)	103	✓	✓						
Zhou et al. [70](A)	926	✓	✓			✓			
Martin et al. [71](A)	118,587	✓	✓						
Awe et al. [72](A)	43					✓			
Jara et al. [34](C)	80	✓	✓	✓		✓			✓
Yu et al. [73](C)	1,628	✓	✓			✓			✓
Goin et al. [74](C)	294	✓					✓		
Ojukwu et al. [75](A)	143	✓				✓		✓	
Ashrap et al. [76](A)	682	✓				✓		✓	
Hamilton et al. [77](A)	23,728	✓		✓					
Basile Ibrahim et al. [78](A)	540,711	✓							
Hardeman et al. [79](A)	1,059	✓		✓					
Lewis et al. [80](A)	474		✓		✓	✓			
Merkt et al. [81](A)	3,747	✓							
Oldenburg et al. [82](A)	386	✓	✓	✓					
Patel et al. [83](A)	103,608	✓	✓				✓		
Bloom et al. [84](A)	310	✓	✓			✓			✓
Zhang et al. [85](A)	8,949	✓	✓		✓				
Theiler et al. [86](A)	2,002	✓	✓	✓					
Park et al. [27](B)	573,634	✓	✓						
Piekos et al. [87](A)	18,638		✓	✓					
Callahan et al. [88](A)	657		✓				✓		
Son et al. [89](A)	546,441		✓		✓				
Fruh et al. [90](A)	108		✓			✓			
Razzaghi et al. [91](A)	110,925	✓	✓						
LEONARD et al. [92](A)	3,308,554	✓	✓						
Geron et al. [93](A)	382		✓		✓				
Clapp et al. [94](A)	36,969	✓				✓			
Wang et al. [95](A)	2,442	✓	✓		✓				
Saadat et al. [96](A)	187		✓					✓	
Barrett et al. [97](A)	1,018	✓	✓					✓	
Gardener et al. [98](A)	485	✓	✓						
Choi et al. [99](A)	171	✓				✓			
Martin et al. [28](B)	90	✓	✓	✓					

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Table 2 (*continued*)

Publications	Sample Size	Demographic and Socioeconomic	Maternal Health and Lifestyle	Pregnancy-Related	Environmental and Community	Biological and Clinical Measures	Health Care and Policy	Psychosocial and Behavioral Measures	Infant and Child Health Factors
Jacobson et al. [35](C)	1,828	✓	✓	✓		✓			✓
Chu et al. [100](C)	2,514	✓	✓		✓				
Aung et al. [101](A)	517	✓	✓						
Oh et al. [102](A)	595	✓	✓						
Varde et al. [103](A)	52		✓			✓			
Grob et al. [104](A)	36,861	✓		✓					
Thorsen et al. [105](A)	56,117	✓	✓						
Ikenoue et al. [106](A)	137	✓	✓			✓			
Oyebode et al. [107](A)	566	✓	✓		✓				
Panchal et al. [108](A)	18,067,310	✓		✓					
Cersonsky et al. [109](A)	3,982		✓				✓		
Alfred et al. [110](A)	10,774	✓					✓		
Stephenson et al. [111](A)	57	✓				✓			
Redhunt et al. [112](A)	510	✓	✓						
Lundeberg et al. [113](A)	36,000	✓		✓					
Boyer et al. [114](A)	8,491	✓	✓						
Eatman et al. [115](A)	247		✓			✓			
Holt et al. [116](A)	528	✓		✓					
Lin et al. [117](A)	1,040	✓	✓			✓			
Brown et al. [118](B)	5,819	✓	✓	✓				✓	
Guan et al. [26](B)	365	✓	✓	✓		✓			
Bartal et al. [119](B)	1,127	✓	✓						
Kanichy et al. [31](C)	29,845	✓	✓	✓			✓	✓	
Sarullo et al. [32](C)	395	✓	✓	✓		✓		✓	✓
Hernandez-Castro et al. [33](C)	421	✓	✓	✓		✓			✓
Chavez et al. [36](D)	3,616	✓	✓		✓		✓		✓
DePaoli Taylor et al. [120](A)	29,821	✓	✓	✓					
Hebert et al. [121](A)	13,372	✓	✓						
Liu et al. [122](A)	162,576	✓	✓	✓	✓		✓		
Lugo-Candelas et al. [123](C)	794	✓	✓						✓
Stanek et al. [124](A)	1,537	✓			✓			✓	
Choi et al. [125](A)	18,236,481	✓							✓
Eick et al. [126](C)	348	✓	✓	✓	✓				
Akapo et al. [127](A)	37	✓					✓		
Kucirka et al. [128](A)	183,640	✓				✓	✓		
Mane et al. [129](C)	109		✓					✓	

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Table 2 (*continued*)

Publications	Sample Size	Demographic and Socioeconomic	Maternal Health and Lifestyle	Pregnancy-Related	Environmental and Community	Biological and Clinical Measures	Health Care and Policy	Psychosocial and Behavioral Measures	Infant and Child Health Factors
Schmidt et al. [130](A)	1,849	✓	✓	✓				✓	
Bodnar et al. [131](A)	10,023	✓	✓					✓	
Sanders et al. [132](B)	236	✓	✓	✓					
Neo et al. [133](C)	11,486	✓	✓	✓					
Pham et al. [134](A)	506,794	✓					✓		
Bommarito et al. [135](A)	900	✓	✓	✓	✓				
Taibl et al. [136](C)	340	✓	✓	✓	✓	✓			
Fabricant et al. [137](A)	4,212					✓			
Barrozo et al. [138](A)	14		✓			✓			✓
Piekos et al. [139](A)	106,428	✓	✓	✓					
Lee et al. [140](A)	2,299	✓		✓		✓			
Mead et al. [141](A)	115	✓	✓	✓		✓			
Fuller et al. [142](A)	68,918	✓		✓			✓		
Hebert et al. [143](A)	1,805	✓	✓	✓					
Ren et al. [144](C)	266,687	✓	✓	✓			✓		
Amorim et al. [145](B)	15,455	✓	✓	✓					✓
Auxier et al. [146](A)	20		✓				✓	✓	
Parry et al. [147](A)	613	✓	✓	✓			✓		
Nguyen et al. [21](A)	56,400,097	✓		✓				✓	
Quintero et al. [148](C)	116,132	✓	✓	✓					✓
Early et al. [149](A)	5,401,899	✓		✓			✓		
Chandler-Laney et al. [150](A)	208	✓	✓	✓		✓			
Mallampati et al. [151](A)	73,596	✓		✓		✓			
Song et al. [152](A)	321	✓	✓	✓		✓			✓
Weber et al. [153](A)	35,498	✓		✓		✓			
Heo et al. [154](A)	1,899	✓	✓						
Suplee et al. [155](C)	2,363		✓			✓	✓	✓	
Poliektov et al. [156](A)	33	✓	✓	✓		✓			
Shara et al. [157](A)	604	✓	✓			✓			
Parikh et al. [158](A)	17,573	✓	✓	✓			✓	✓	
Rose et al. [159](A)	50	✓	✓			✓			
Simeone et al. [160](C)	12,879	✓		✓			✓		
Barcelona et al. [161](A)	1,117							✓	
Trasande et al. [162](C)	3,619	✓	✓	✓	✓				✓
Mancuso et al. [163](A)	179	✓	✓			✓		✓	
Wright et al. [164](A)	335	✓		✓	✓	✓		✓	

(*continued on next page*)

Table 2 (*continued*)

Publications	Sample Size	Demographic and Socioeconomic	Maternal Health and Lifestyle	Pregnancy-Related	Environmental and Community	Biological and Clinical Measures	Health Care and Policy	Psychosocial and Behavioral Measures	Infant and Child Health Factors
Hernandez-Castro et al. [165][(A)]	329	✓		✓	✓	✓			✓
Campbell et al. [166][(B)]	14,967,861	✓	✓	✓			✓		
Roese et al. [167][(A)]	325,581	✓	✓	✓	✓	✓	✓		

[(A)]: Pregnant Cohort
[(B)]: Postpartum Cohort
[(C)]: Pregnant Women and Infants
[(D)]: Postpartum and Infants
Demographic and Socioeconomic Factors: Maternal demographics (age, marital status, education, SES, race/ethnicity, employment, insurance, income, residence, nativity), length of US residence.
Maternal Health and Lifestyle Factors: BMI, smoking (pre/during pregnancy), alcohol, drug use, pre-existing conditions, nutrient levels, allergen sensitization, mental health (depression, anxiety), sleep, exercise, and diet.
Pregnancy-Related Factors: Parity, gestational age, birth weight, mode of delivery, pregnancy complications, breastfeeding, ART use, pregnancy intention, vaccination status.
Infant and Child Health Factors: Infant gender, allergen sensitization, neonatal outcomes, childhood diseases.
Environmental and Community Factors: Air and soil pollution, housing conditions, community state type, neighborhood poverty, access to healthy food, climate.
Biological and Clinical Measures: Biomarkers, blood/urine analytes, genetic markers, inflammation markers, allostatic load.
Health Care and Policy Factors: Insurance type, government program participation, prenatal care, state policies, hospital characteristics.
Psychosocial and Behavioral Measures: Discrimination experiences, resilience, social support, stress/trauma exposure, substance use.

and address maternal health disparities.

3.5.1.1. Outcomes inference and results validity. In assessing maternal health disparities, odds ratios (ORs) and confidence intervals (CIs) have been frequently used as a statistical method, used in 38 and 29 studies respectively. For example, research conducted by Goyal et al. and Kitsantas et al. [25,37] rely mainly on ORs and CIs to analyze maternal health data. Adjusted odds ratios (AORs) are also important, as illustrated by Dunlop et al. and McQueen et al. [38,47].

Statistical significance analysis, often assessed through p-values, is another prevalent approach to determining the reliability of findings. This method was predominantly used in studies by Gillespie et al. and Johnson et al. [50,66]. Additionally, relative risks and hazard ratios provide information on the likelihood of outcomes, effectively employed in Creanga et al. and Bruckner et al. [43,49], respectively. The Area Under the Receiver Operating Characteristic (AUC) curve is highlighted by Menon et al. and Weber et al. [44,61].

Regression and beta coefficients, which measure the influence of predictors on outcomes, were utilized to conclude maternal health disparities, as evidenced in studies conducted by Barrett et al. and Gardener et al. [97,98].

Several robust methodologies ensure the validity of findings on maternal health disparities. Seventeen studies frequently employ standardized statistical methods and control for multiple tests to enhance reliability, as exemplified by Holm's procedure in Goyal et al. [25]. Large, diverse datasets provide a solid foundation for generalizability, improving external validity, as seen in studies by Bruckner et al. and Johnson et al. [49,66].

Regarding validating methods, Dolbier et al. [42] used the Edinburgh Postnatal Depression Scale (EPD) to conduct reliable psychological assessments. Validated laboratory procedures for chemical measurements, as demonstrated in the studies by Bloom et al. and Aung et al. [84,101], accurately capture biophysical data, improving the interpretability of results.

3.5.2. Machine learning studies

Among the 147 studies reviewed, 18 employed ML techniques. These studies focused on predicting outcomes such as spontaneous preterm birth (16.67 %), early spontaneous preterm birth in specific racial groups (11.11 %), and the impact of maternal life events and stress on birth outcomes (16.67 %). Additionally, ML was used to explore the effects of maternal SARS-CoV-2 infection (11.11 %), identify urban environmental factors influencing prenatal care (11.11 %), and reduce bias in predicting postpartum depression (27.78 %). Neural networks were utilized to examine how socioeconomic factors and race affect fetal development (5.55 %). Natural language processing (NLP) was applied to forecast severe maternal morbidity and childbirth-related PTSD. Integrating ML into maternal health research enhances outcomes for mothers and infants, improves prediction accuracy, uncovers complex relationships, and facilitates targeted interventions and early detection. Fig. 4 illustrates the count of ML algorithms utilized across the reviewed studies.

Hierarchical clustering in the study by Zhang et al. [85] and multivariate adaptive regression splines (MARS) in Menon et al. [44] identify subgroups and non-linear relationships in maternal health factors. Random forests (RF) discussed in Piekos et al. [87] and extreme gradient boosting (XGB) algorithms outlined in Park et al. [27] offer robust classification and prediction abilities, effectively handling diverse data types with high accuracy. Other methods like lasso and ridge regression, the study conducted by Weber et al. [61] plays a role in selecting features for predictive modeling. They analyze datasets combining institutional factors like health insurance and medical history with social factors like education and ethnicity to pinpoint significant predictors of early spontaneous preterm births (sPTB).

3.5.2.1. Optimization and evaluation of machine learning models in maternal health disparity studies. Menon et al. [44] used the MARS algorithm to select features, while Clapp et al. [94] incorporated only features exceeding certain frequency thresholds to focus on the most informative predictors. Cross-validation strategies varied, with some studies, such as Bartal et al. [119], using 10-fold cross-validation to verify model stability, while others, such as Park et al. [27], splitting data into training and test sets to evaluate model performance.

Model evaluation metrics, such as AUC and calibration plots, measured model accuracy and reliability, as demonstrated by Clapp et al. [94]. Handling imbalanced data, a common challenge in healthcare datasets, was addressed using techniques such as downsampling in Bartal et al. [119] and the Synthetic Minority Over-sampling Technique (SMOTE) in Cersonsky et al. [109], ensuring that the models were robust against biased data representations.

Table 3
Categorized Studies on Maternal Health Disparity.

Articles	Focus	Inclusion/ Exclusion Criteria	Outcomes
Racial/Ethnic and Socioeconomic Disparities			
Kitsantas et al. [37]	Breastfeeding and BMI disparities in diverse, low-middle SES women	Study of singleton births excluding specific demographics	SES and BMI impact breastfeeding rates among Hispanic and Black women
Wallace et al. [41]	Examines how pre-pregnancy stress and race, alongside neighborhood poverty, affect birth outcomes	Louisiana BHS births 1990–2009 included	Louisiana BHS: African American women, stress, poverty, no birth outcome link, racial disparities
Creanga et al. [43]	US study: Racial/ ethnic differences in severe maternal complications during childbirth	Severe maternal complications by race/ethnicity in large states, excluding select hospitals	Severe maternal morbidity: Non-Hispanic Black women, age, insurance, SES, chronic illnesses
Menon et al. [44]	Preterm birth prediction models, biomarker variations: African American, Caucasian women	Pregnant women aged 18–40, excluding specific conditions	Biomarkers vary in predicting preterm birth for African American vs. Caucasian women
Drassinower et al. [45]	Maternal race impact on twin pregnancy outcomes, including delivery methods and disparities	Twin gestations ≥ 32 weeks, vertex-vertex, excluding certain conditions, missing race data	Cesarean rate disparities: African Americans fewer electives, Asians more; similar outcomes
Stephenson et al. [48]	Assessing labor outcomes in prostaglandin-induced labor among diverse racial/ethnic groups for disparities	Exclusion criteria: Contraindications, urgent delivery, misoprostol/ dinoprostone sensitivity	Black women: more cesareans, smaller babies; Hispanic women: higher postpartum hemorrhage
Zamora-Kapoor et al. [53]	Studying pre-eclampsia risk in AI/ AN versus White women, factoring in obesity (BMI) affects	Inclusion: Singleton first-time mothers; Exclusion: Multiple gestations, prior live births	AI/AN women had higher pre-eclampsia risk than White women, but BMI adjustment equalized it
Lemon et al. [54]	To assess pre-pregnancy obesity's role in Black-White stillbirth and infant mortality disparity	Included singleton births to non-Hispanic Black or White women, excluding some instances	Maternal obesity explains 10 % of racial disparity in stillbirth and infant mortality; severe obesity, 5 %
Giurgescu et al. [23]	Investigating the influence of racial bias on systemic inflammation in expectant African American mothers	Adults ≥ 18, low-risk pregnancies included. Excluded: Medical complications, steroids	Racial discrimination linked to higher IL-4, IL-6; no other significant associations
Hill et al. [55]	Assessing how higher progesterone levels later contribute to racial differences in live birth outcomes	Fresh autologous embryos. Excluded: Donor oocytes, frozen transfers, specific screening	High progesterone reduces live births in all races, persisting disparities in minorities post-adjustment
Weber et al. [61]	ML predicts early preterm births in black and white women from a large cohort	Including singleton births to nulliparous black and white women, excluding missing data	Predicting early preterm birth was tough, and factors varied by race
LEONARD et al. [92]	Evaluate comorbidity scoring in diverse groups, impact on maternal morbidity disparities	Individuals who had live births recorded in California from 2011 to 2017	Persistent maternal morbidity disparities despite comorbidity scoring adjustments
Redhunt et al. [112]	Examining racial disparities in preterm birth, considering resilience's impact	Eligible: Pregnant, 18+, non-Hispanic, completed 75 %+ CD-RISC survey, with re-enrollment	Resilience did not alter racial preterm birth gaps; Black rates stayed higher than White rates
Holt et al. [116]	Study: Maternal care, racial disparities, outcomes for women of color, including delivery type, anesthesia	Included maternal event incident reports matched with deliveries, excluding duplicates	Non-Hispanic Black overrepresented, tied to cesarean deliveries and specific anesthesia
Bruckner et al. [49]	Hypothesis: First-born sons raise preterm delivery risk in diverse groups, as in Scandinavia	Mothers with singleton live births for their first and second children, excluding specific cases	After a first-born male, preterm birth risk increases except in non-Hispanic blacks
Lewis et al. [80]	Studies of water TTHM levels, spontaneous preterm birth, and related factors	The participants included individuals residing in Philadelphia with available TTHM data	There is no link between TTHM and preterm birth; race-TTHM interaction affects beta-defensin-2 levels
Aung et al. [101]	Examines maternal PFAS and pregnancy depression, comparing US-born and immigrant women	Included English or Spanish-speaking pregnant participants 18 + with a singleton fetus	PFAS linked to higher depression scores, especially in immigrant women
Lin et al. [117]	Study on trace elements, first-trimester maternal kidney function, with race/ethnicity and socioeconomic status	Only pregnant women with complete first-trimester data were included, excluding others	Lead lowers kidney function, while magnesium, selenium, and zinc improve it
Miller et al. [57]	Studies the impact of maternal socioeconomic status on fetal development and evaluates Centering-Pregnancy efficacy	Included pregnant women aged 18 or older with specific criteria, excluding certain conditions	Maternal disadvantage affects gene expression, with Centering-Pregnancy mitigating some changes
Yusuf et al. [65]	Compares high-strength folic acid with standard doses in promoting prenatal brain growth	Examined pregnant smokers aged 18–44 near Tampa, FL, up to 21 weeks gestation.	High folic acid preserved BBR, while elevated cotinine impaired fetal brain development
Gerson et al. [68]	Aims to study how cervicovaginal microbiota and a short cervix jointly affect the risk of spontaneous preterm birth.	Included women at a specific hospital for prenatal care, excluding specific cases	Cervicovaginal CST IV and a short cervix both increase sPTB risk, especially together
Basile Ibrahim et al. [78]	Analyzes VBAC rates in the United States and factors affecting outcomes	Included women with 1–2 prior cesarean deliveries, excluded those with missing data	Study shows VBAC rates are higher with Medicaid/ self-pay, lower in the South, and higher in the Midwest than Northeast
Thorsen et al. [105]	Analyze predictors, factors, and disparities related to obstetric bypassing	Montana resident hospital births 2014–2018, excluding non-	Reservation and AI/AN individuals traveled further for obstetric care,

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Table 3 (*continued*)

Articles	Focus	Inclusion/ Exclusion Criteria	Outcomes
	and travel distance for care	residents and incomplete records	highlighting rural access disparities
Oyebode et al. [107]	Study PM2.5 pollution's link to CST IV and racial disparities in CST IV prevalence	People with cervicovaginal microbiota, addresses, and ≥ 1 month of early pregnancy PM2.5 data	PM2.5 exposure correlates with higher CST IV rates, especially among Black individuals
Martin et al. [71]	Unmet SUD treatment needs in U.S. reproductive-age women, including pregnant, and sociodemographic factors	Included women aged 18–44 in the analysis, with specific exclusion criteria not provided	Reproductive-age women face SUD treatment barriers due to racial and socioeconomic factors
Panchal et al. [108]	Study pregnancies with retained IUDs: characteristics, incidence, factors, delivery outcomes	Included hospital-based deliveries (2016–2020) for women aged 15–54, no specified exclusions	Retained IUDs, rare in hospital births, are linked to adverse outcomes in specific demographics
Alfred et al. [110]	Examine disparities in OBGYN perioperative care, considering changes since March 2020	Included females aged 18 + admitted perioperatively under OBGYN faculty	Post-March 2020, Black patients' noncompliance varied, notably in BSGS, sans overall racial disparity
Stephenson et al. [111]	Study preeclampsia markers, lipid profiles, and birth timing, with racial variations	–-	Identify lipid predictors for early preeclampsia, assess racial variations, and evaluate risk stratification
Patel et al. [83]	Analyzes prenatal substance use screening rates by substance type, year, state, and screening influencers	Survey included recent live birth women from 2016 to 2018 PRAMS; exclusions not stated	The study found screening rates varied by substance, state, and demographics
Geron et al. [93]	Metal exposure in pregnant women, factors, and risks	Exclude: <18, HIV+, >7 drinks/ week pre-pregnancy, non-English/Spanish speakers, multiples	Metal exposure disparities in pregnant women tied to lower socioeconomic areas
Wang et al. [95]	Study looks at early pregnancy PBDE exposure and its effect on delivery timing, factoring in demographics	Women 18–40, singleton pregnancy, 8–13 weeks, no significant conditions, with or without obesity	Higher PBDE 153 levels shorten gestational age in non-obese but not obese women
Koning and Ehrenthal [64]	Studying how race, income, and maternal life events affect birth outcomes via stressor mapping	The study included respondents from the PRAMS dataset	Three stressors affect minority and low-income groups' birth outcomes
Carroll et al. [69]	Studying sleep's influence on gene expression, racial inflammation gaps, and pregnancy outcomes	Pregnant women under 30 weeks at OSUWMC, excluding health conditions and drug use	Sleep worsens inflammation in pregnant African American women
Hardeman et al. [79]	Analyzing how police contact affects preterm birth risk across diverse racial groups and its role in disparities	Pregnant individuals delivering singletons at certain facilities, with exclusions	Police encounters increased preterm birth risk, revealing racial disparities

Table 3 (*continued*)

Articles	Focus	Inclusion/ Exclusion Criteria	Outcomes
Callahan et al. [88]	Studying pregnant women's vaccine views and evidence-based information influence	The study focused on pregnant clinic patients, excluding incomplete surveys	Racial and education gaps affect vaccine acceptance in pregnant women, tied to pre-pregnancy habits
Eatman et al. [115]	Studying how phthalates, BPA, and stress affect birth outcomes in expectant African American women	US-born pregnant Black women, 18–40, with no chronic conditions; excluded others	Phthalate/BPA exposure links to lower birthweight with stress, no effect on gestational age
Hansen et al. [51]	Study predicts success in unexplained infertility with IUI and ovarian stimulation, assessing drug effectiveness	Included women aged 18–40 with specific criteria, without mentioning exclusion criteria	Age, waist size, income, infertility duration, and prior pregnancy loss predict outcomes
Clapp et al. [94]	Use ML and NLP on admission notes to predict maternal morbidity risk without extra patient data	Included all encounters involving delivery but did not specify exclusion criteria	AUC > 0.70 for SMM and nt-SMM. NLP matches EOCS for early high-risk patient detection
Choi et al. [99]	Aims to study melamine exposure in U.S. pregnant women with participant characteristics and samples	Inclusion criteria: At least 6 mL of urine collected during pregnancy	Melamine, aromatic amine exposure in US pregnant women; racial disparities; cotinine link
Kanichy et al. [31]	Examines racial gaps in North Dakota breastfeeding, focusing on how interpersonal violence affects American Indian women	Inclusion criteria were recent mothers from live birth certificates, excluding those with incomplete survey data.	American Indian women had lower breastfeeding rates than White women, unrelated to interpersonal violence.
Jacobson et al. [35]	Examines fetal growth patterns concerning maternal factors and health, assessing variations by maternal race/ ethnicity	Study included women with live births (2016–2019), excluding twins, missing ultrasounds, and congenital anomalies.	Fetal size varies by race, with Black, Hispanic, and Asian fetuses smaller than White, notably in US-born Asians.
Yu et al. [73]	Examines maternal immune activity during pregnancy, child self-regulation, and socio-economic/ racial differences in behavior	Study included pregnant women giving blood samples and children assessed at age 4; no exclusions were specified.	SES disparities and maternal IL-8 levels affected child self-regulation; no racial disparities were significant.
Hoepner al. [29]	Focuses on BPA levels and predictors in an urban birth cohort	Included mothers with prenatal BPA; children tested at 3, 5, or 7; excluded smokers and certain health issues.	High BPA detection across ages, with race, marital status, and season variations and correlations with phthalate levels.
Gillespie et al. [58]	Analyzing childhood stress effects on birth timing in African American women, isolating maternal cortisol	African American adults, 18–34 at conception, with prenatal ultrasounds	Childhood stress predicts birth timing, is influenced by cortisol, and varies in impact across subgroups
Bartal et al. [119]	Study on childbirth narratives for CB-	Included: Women ≥ 18, gave birth	ML model effectively

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Table 3 (*continued*)

Articles	Focus	Inclusion/ Exclusion Criteria	Outcomes
	PTSD detection using NLP/ML	within 6 months. Excluded: <30-word narratives, unspecified criteria.	identified CB-PTSD based on word count, negativity, and specific word types in narratives
Siegel et al. [67]	Study on automated postpartum follow-up within 72 h and factors related to postpartum concerns	Included women aged 18 + giving birth at UCSF, with undisclosed exclusion criteria	High response rates to automated calls reveal race, insurance, and primiparous disparities.
Sarullo et al. [32]	Examines how SES, race, and maternal adversity affect fetal development, guiding interventions/policy	Included: drug-free participants (except cannabis) without pregnancy issues; excluded: those with brain injuries.	Income and neighborhood factors better predicted birth weight than race, which didn't impact other outcomes.
Guan et al. [26]	Study patient characteristics in severe PPH interventions and racial care disparities	Included: PPH with transfusion; Excluded: No antenatal visits, incorrect gestational age	Black patients receive fewer postpartum hemorrhage interventions than Whites, revealing disparities.
DePaoli Taylor et al. [120]	Investigate gonorrhea's link to hypertensive disorders, preterm birth, and preeclampsia subtypes.	Included: Singleton pregnancies at the University of Texas Medical Branch; excluded: missing data.	Gonorrhea increases preterm risks but not hypertensive pregnancy disorders.
Liu et al. [122]	Examine racial disparities in severe maternal morbidity during COVID-19 and the impact of SARS-CoV-2 infection.	Included: Live births in South Carolina (Jan 2018–June 2021); excluded: terminations, stillbirths, and missing data.	SMM rose 12.7 % in the pandemic, with persistent racial disparities and higher risks linked to SARS-CoV-2.
Wright et al. [164]	Examine the impact of racial composition and discrimination on inflammation and pregnancy outcomes.	Inclusion: Black or African American, singleton pregnancy, age 18–45, English-speaking/reading; Exclusion: Multiple gestations.	Racial factors and discrimination raised MIF, tied to preterm birth risk.
Neo et al. [133]	Investigate neighborhood socioeconomic position and its impact on gastroschisis, maternal age, and BMI.	Included: Singleton case infants with gastroschisis and non-malformed controls; Excluded: Multiple births or incomplete data.	Lower nSEP increased gastroschisis risk, with maternal age and neighborhood conditions influencing risk patterns.
Ren et al. [144]	Develop ML models for low birthweight prediction and analyze the importance of features.	Inclusion: Maternal delivery claims; Exclusion: Abortion, false labor, and missing linkage.	Using maternal race, age, and health, ML predicts low birth weight with rebalancing.
Nguyen et al. [21]	Measure state-level racial sentiments on Twitter and their link to preterm birth and low birth weight.	Included U.S. tweets with race-related keywords, focusing on singleton births without abnormalities.	Negative racial sentiment is linked to higher adverse birth outcomes, especially in racially minoritized groups.
Early et al. [149]	Analyze adverse pregnancy outcomes in SCD patients and racial disparities in outcomes.	Inclusion: Deliveries for ages 11–55; Exclusion: Missing data, miscarriage, multifetal gestation.	SCD pregnancies have higher severe maternal morbidity and mortality, with racial disparities increasing risks.
Parikh et al. [158]	Examine the link between lifestyle, socio-demographics, GDM risk, and healthcare affordability for pregnant women.	Inclusion: Pregnant women, valid BMI (15–50), complete records; Exclusion: Missing or unreasonable values (e.g., extreme outliers).	Higher BMI, depression, and lower fruit intake raise GDM risk, with socioeconomic factors affecting affordability.
Heo et al. [154]	Investigate exposure misclassification from residence at delivery and the impact of residential mobility on pregnancy exposure estimates.	Inclusion: Pregnant women ≥ 18, <18 weeks, non-medically threatened, planning delivery at study hospitals.	Exposure misclassification, influenced by mobility and socioeconomic factors, led to overestimations of PM2.5 and EVI.
Disparities in Birth Outcomes and Pregnancy Complications			
Dunlop et al. [38]	Study linking nutrients, preterm birth, and racial disparities in U.S. women	Study of adult women with singleton pregnancies, excluding certain complications	Link between dietary fats, preterm birth, and vitamin D disparities by race
Wegienka et al. [40]	Disparities in allergies and asthma between black and white pregnant women	Study of WHEALS cohort excluding financially disadvantaged pregnant women	Black women had higher IgE levels and eczema likelihood, disparities persisting after adjustments
Gillespie et al. [50]	Examining pregnancy and postpartum immune response changes to explore racial adaptation disparities	Excluded multiple fetuses, anomalies, chronic conditions, specific drugs, and others	Mid-pregnancy: Higher TNF-alpha in Black women, slightly lower postpartum IL-1 beta
Hamilton et al. [77]	Analyzes military healthcare's effect on maternal morbidity rates by race, detecting disparities	Used data from large MTFs, focusing on Black and White women, excluding other races	Racial health disparities persist, except for postpartum hemorrhage requiring transfusion
Zahran et al. [24]	Study links soil-led exposure to eclampsia risk pre- and post-Hurricanes Katrina and Rita	Included mothers from New Orleans with zip code info, no detailed exclusions	Eclampsia risk is higher with soil lead, lower where it decreased post-Katrina and Rita
Chang et al. [46]	Analyzing postpartum HBV care follow-up rates and related factors	Included women with positive HBsAg, excluding those with prior HBV specialty care	HBV + moms lacked postpartum follow-up; primary care provider and HBeAg positivity
Johnson et al. [66]	Aims to study pain disparities by race, analyze pain scores, and post-cesarean birth medication	Cesarean births with live neonates, excluding specific cases	Minority women had more pain but got less care than White women, even after clinical adjustments
Martin al. [28]	Study on early maternal warnings and postpartum severe outcomes prediction	Women 12–55, 80 %+ live birth enrollment: postpartum checks, mental health	Maternal warning links to SMMM. Risks: Black race, higher BMI, cesarean; no racial effect
Disparities Related to Chemical and Environmental Exposures			
Unal et al. [22]	Study on racial differences in BPA levels between mothers and fetuses	Study on healthy, first-time mothers with term pregnancies	Higher BPA levels in African American mothers

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Table 3 (*continued*)

Articles	Focus	Inclusion/ Exclusion Criteria	Outcomes
			and Hispanic fetuses
Christian et al. [39]	Impact of EBV and discrimination on pregnancy, varying by race	Excluded those with specific medical conditions during pregnancy	Higher EBV antibodies in African American women linked to racial discrimination
Most et al. [59]	Determining factors related to energy balance during early pregnancy in African American and white women	Obese women (BMI ≥ 30), 18–40, singleton pregnancies. Excluded: smokers and substance	African American women may overestimate energy needs due to lower expenditure than white
Patten et al. [62]	Evaluates an intervention's effect on tobacco use in pregnant Alaska Native women.	Included Alaska Native women 18 + from villages with phone access, some exclusions	High tobacco uses in pregnant Alaska Native women, primarily Iqmik; most are ready to quit
Agrawala et al. [63]	Looking at how preconception low-dose aspirin impacts pregnancy outcomes, considering socioeconomic status.	Women 18–40, regular cycles, 1–2 prior losses, no aspirin contraindications, or primary conditions	LDA boosted birth and pregnancy rates across income and education, with no mid-SES pregnancy loss impact
Piekos et al. [87]	The study examines the effects of SARS-CoV-2 on birth outcomes and gestational age after infection	Pregnant 18–45 y/o, single pregnancies, specific insurance, excluded multiples or different insurance	SARS-CoV-2 heightens preterm and stillbirth risks, less so symptom severity's link to delivery time
Fruh et al. [90]	Analyzes pregnant women's product use and urinary phthalates for exposure sources during pregnancy	Pregnant women, 18+, under 15 weeks, no preexisting diabetes or multiples, at BIDMC	Certain personal care products are linked to higher phthalate levels, with racial exposure differences
Ashrap et al. [76]	Studying how psychosocial stress, mental exposure, and lower status impact birth outcomes	Women aged 18–40 planning delivery in Northern Karst aquifer region hospitals, with exclusions	Stressed women's birth outcomes emphasize metal exposure effects
Jara et al. [34]	Investigates pregnancy adiponectin and leptin levels, weight gain, infant birth weight, and the influence of race/ BMI	Included OSUMC pregnant women; excluded hypertension, diabetes, immune issues, anomalies, and substance abuse.	Leptin influenced weight gain but not birth weight, with racial disparities in leptin/adiponectin among Black women with obesity.
Goin et al. [74]	Examines prenatal environmental chemicals' impact on maternal and infant health, addressing socio-demographic disparities	Participants spoke English or Spanish, were over 18, and had a singleton pregnancy at 13–27 weeks gestation.	Chemical exposures varied by hospital, income, and race, with higher levels in low-income groups.
Bloom et al. [84]	Gestational phthalate exposure effects on neonatal HC and CI, considering race and sex	Singleton pregnancies, age 18+, planning MUSC delivery, excluding fetal sex preferences	Third-trimester MEP exposure linked to smaller newborn head size in white mothers and girls
Stanek et al. [124]	Assess lead exposure in pregnant women through biological and residential measures and explore related demographic factors.	Pregnant women who provided blood, urine, and/ or surface wipe samples during pregnancy and postpartum for lead analysis.	Lead exposure in pregnant women correlates with race, income, and residential lead levels, showing demographic disparities.
Eick et al. [126]	Examine the joint effects of PFAS and stress on birth outcomes and health disparities.	Inclusion: African American females, U.S. born, fluent in English, no chronic conditions. Exclusion: Others not meeting criteria.	Exposure to PFAS and stressors lowered birthweight z-scores but did not affect gestational age at delivery.
Trasande et al. [162]	Evaluate prenatal phenol and paraben exposure on birth outcomes, trimester effects, and racial/ ethnic disparities.	Inclusion: Pregnant individuals with phenol data, gestational age, and singleton deliveries; Exclusion: Cohorts < 50 or incomplete data.	Certain chemicals lowered birth weight; exposure was higher in Hispanic and Black mothers.
Hernandez-Castro et al. [165]	Evaluate prenatal OPE exposures and their impact on infant motor development, including sex-specific effects.	Inclusion: Pregnant women < 30 weeks, ≥18, fluent in English/Spanish; Exclusion: Multiple gestations, disabilities, incarceration, HIV + .	Prenatal DPHP exposure raised fine motor delay odds by 26 %, with female infants more affected by BCIPP and DPRP.
Bommarito et al. [135]	Examine trends in phthalates and phenols, predictors, and exposure clusters in pregnant individuals.	Inclusion: Singleton live births, LIFECODES cohort, and available urine samples; Exclusion: Incomplete birth outcome data.	Phthalates declined, replacements and BPS rose, with exposure disparities by race, BMI, and insurance.
Taibl et al. [136]	Assess trimester-specific PFAS effects on birth outcomes and biomarker confounding.	Inclusion: African American women, 18–40, singleton pregnancy; Exclusion: Multiple pregnancies, chronic conditions.	PFAS affected birthweight differently by trimester, with robust results across analyses.
Disparities in Healthcare Access and Treatment			
McQueen et al. [47]	IVF outcomes: Assessing race, identifying disparities in success rates	First-time autologous fresh IVF cycles, excluding specific types and multiple/ unknown races	IVF racial disparities: Lower success for Black, Asian women; higher spontaneous abortion rates
Zhang et al. [85]	Built environment's impact on urban women's health, including PPD	Pregnant women aged 18–45 with live deliveries, excluding outliers	Prenatal care ties PPD rates to building environment factors like retail ratio and pollution
Cersonsky et al. [109]	Build stillbirth prediction models, refining with pregnancy data and assessing variable importance	Singleton pregnancies with enough data, resulting in stillbirth or live birth during the study	ML predicted stillbirth risk with 85 % AUROC from demographics and prenatal metrics
Lundeberg et al. [113]	Assess if Military healthcare equalizes SMM and preeclampsia disparities	TRICARE beneficiaries birthing at a Military Treatment Facility, April 2019 – March 2020	TRICARE's maternal morbidity disparities: Black/ API women face higher preeclampsia risks

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Table 3 (*continued*)

Articles	Focus	Inclusion/ Exclusion Criteria	Outcomes
Bloom et al. (Bloom et al., 2019)	Investigates how maternal phthalate exposure affects birth outcomes, considering race and infant sex	Included: Women ≥ 18, singleton, MUSC delivery. Excluded: anomalies, steroids, diabetes, endocrine issues.	MEP exposure ties to smaller head size, stronger in white mothers and females; no cephalization link.
Hsu et al. [56]	Aims to study ectopic pregnancy care and treatment choices	Included women aged 15–60 with ectopic pregnancy without specifying exclusion criteria	2006–2015: More methotrexate, less surgery; racial/ insurance gaps; fewer issues
Brown et al. [118]	Study on postpartum program impact and demographic disparities in hospitalizations and maternal morbidity	Inclusion: High-risk women based on ICD-10. Exclusion: Unspecified	MOMs program reduced rehospitalization, lessening racial disparities in postpartum care
Chavez et al. [36]	Study links social needs to postpartum depression, examines higher PPD risk with social needs, and details symptom prevalence.	Included: Mothers with timely EPDS and social screens. Excluded: non-mothers, infants > 6 months, delayed screens.	Social needs are linked to higher PPD odds, more symptoms, and varied risks by demographics.
Hernandez-Castro et al. [33]	Studies prenatal OPE exposure effects on gestational age and birth weight, considering sex-specific differences	Included: <30 weeks, ≥18, English/Spanish. Excluded: multiples, disabilities, incarceration, HIV.	OPE metabolites showed no overall birth outcome effects, but BDCIPP was linked to earlier births in males.
Campbell et al. [166]	Explore racial/ ethnic disparities in postpartum VTE prophylaxis eligibility and risk factors.	Delivery hospitalizations were included if identified using validated procedures, excluding those with missing race and ethnicity data.	Black postpartum individuals have the highest VTE prophylaxis eligibility due to clinical risk factors, showing disparities.
Choi et al. [125]	Analyze TFAL's impact on out-of-state births and maternal-neonatal health in Tennessee.	Included: Live singletons born to individuals aged 15–44 in states with 2003 Birth Certificate revisions by 2010.	TFAL implementation increased out-of-state births among Tennessee residents by 13 % compared to other states.
Akapo et al. [127]	Assess access, travel, and wait times for OBGYN subspecialists near IHS hospitals for AI/ AN populations.	Inclusion: OBGYN subspecialists near IHS hospitals, AI/ AN women and pediatrics within 60 min; Exclusion: Hawaii/Alaska data.	AI/AN women face long drives to care, with a median of 214 min; 37 % of clinics reject IHS referrals.
Barcelona et al. [161]	Develop NLP methods to identify stigmatizing language in clinical notes and address perinatal health inequities.	Inclusion: Birthing people > 20 weeks' gestation in 2017; Exclusion: Irrelevant clinical notes.	ML identified stigmatizing language linked to bias and inequities in healthcare contexts.
Kucirka et al. [128]	Describe pregnancy-related ESKD characteristics, survival, transplantation access, and nephrology care.	Inclusion: Reproductive-aged women with pregnancy-related ESKD; Exclusion: Missing CMS Form 2728 or cause data.	Black patients faced more pregnancy-related ESKD but had similar survival rates.
Simeone et al. [160]	Investigate maternal COVID-19 experiences, racial/ ethnic differences, and equitable protection efforts.	Inclusion: Women with live births (April–December 2020) who answered COVID-19 questions; Exclusion: Incomplete surveys.	COVID-19 worsened delivery disparities, with more infant separations and support restrictions for minorities.
Mane et al. [129]	Develop Rosie, a new mothers chatbot to address health disparities and engage community input.	Inclusion: Female participants, 14+, pregnant or parenting a child under 3, from a minoritized group.	Mothers highly valued the Rosie chatbot, showing a strong interest in using it for baby health and development information.
Schmidt et al. [130]	Examine ACEs, violence, and racial disparities in prenatal care access for American Indian women.	–	American Indian women reported more late and unsatisfactory prenatal care, unaffected by violence.
Pham et al. [134]	Assess HBsAg testing prevalence, HBV treatment gaps, and factors influencing screening among pregnant persons in the U.S.	Inclusion: Pregnancies aged 15 – 55 with 42 weeks enrollment; Exclusion: HIV or hepatitis C diagnoses before delivery.	HBV care showed racial and educational disparities, with under 50 % postpartum monitoring and better testing for Asians.
Fabricant et al. [137]	Evaluate the PPV of administrative data for severe maternal morbidity with and without the transfusion indicator.	Inclusion: Childbirth admissions at LAC + USC (2016–2019); Exclusion: Pregnancies < 20 weeks and incomplete charts.	Excluding transfusion improves SMM accuracy; preeclampsia is underreported.
Piekos et al. [139]	Evaluate COVID-19 vaccination effects on maternal infections and fetal outcomes during pregnancy.	Inclusion: Pregnant 18–44, singleton > 20 weeks; Exclusion: Prior COVID-19, multiples, <20 weeks.	Vaccination reduced SARS-CoV-2, stillbirth, preterm birth, and low birthweight; boosters improved stillbirth rates.
Hebert et al. [143]	Assess prenatal alcohol counseling differences by race.	The study included AI/AN and NHW women who drank in the 2 years before pregnancy, excluding non-drinkers.	AI/AN women received more counseling, while higher education and income reduced it, highlighting disparities.
Auxier et al. [146]	Examine eHealth self-monitoring adaptation and pregnant users' interactions impacting access and usage.	Inclusion: Pregnant women 18+, second trimester, with smartphone access and Finnish/ English skills.	eHealth engagement varies by personal and systemic factors, highlighting the need for personalized approaches.
Parry et al. [147]	Assess the impact of advanced oral hygiene on maternity outcomes, preterm birth, and health disparities.	Inclusion: Singleton pregnancy, gingivitis, 20 + teeth; Exclusion: Severe oral conditions, multiple gestations, HIV, diabetes, meds.	Advanced oral hygiene may reduce preterm birth risk, with disparities in disadvantaged groups.
Disparities in Maternal and Neonatal Health			
Zhou et al. [70]	Analyzing minority impact on donor oocyte pregnancy, birth rates,	First fresh oocyte donations. Excluded: Genetic testing, missing	Black and Hispanic recipients show lower rates of pregnancy and live

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Table 3 (*continued*)

Articles	Focus	Inclusion/ Exclusion Criteria	Outcomes
	considering age, BMI, infertility diagnosis	race data, or shared donors	births than White recipients
Merkt et al. [81]	Contrasting pregnancy-related mortality rates: urban vs. rural, demographics (race, ethnicity, age)	Women who died within a year of pregnancy due to complications, excluding some cases	Rural areas see higher pregnancy-related mortality with racial/ethnic disparities linked to age
Son et al. [89]	Study on late heat and preterm birth in North Carolina.	Births from May-Sept 2003–2014, gestational age 28–44 weeks, excluding under 400 g or outliers	Preterm birth risk rises with heat in poorer, less green urban areas, excluding heat waves
Grob et al. [104]	Evaluate military healthcare for equitable maternal morbidity rates among races/ ethnicities	Military deliveries included; Alaskan Native/American Indian groups excluded	API women had higher specific SMM and hemorrhage rates, similar overall SMM as others
Oldenburg et al. [82]	Creates genomic inflammation index (GII) as a placental biomarker, explores link to perinatal factors	ELGAN study participants: mothers and infants born before 28 weeks gestation	Black mothers display heightened GII disparities, tied to marital status and steroid use sensitivity
Awe et al. [72]	Studying placental gene axis in term pregnancies for EOSPE and its link to vitamin D	Those < 75 % compliant with supplementation were excluded from the vitamin D analysis	Study links placental gene expression, race/ ethnicity, and vitamin D sufficiency
Ikenoue et al. [106]	Study looks at maternal insulin resistance and fetal fat in mid to late gestation using different methods	Straightforward pregnancies included complicated cases excluded	Mom's insulin resistance at 20- and 30 weeks is linked with baby's fat, regardless of BMI and race
Chu et al. [100]	Analyzes immigrant enclave effects on child birthweight, considering prenatal air pollution and maternal birthplace	Included: Children 0–48 months. Excluded are illness, pre/post-term, smoking, immigration, and missing data.	Foreign-born mothers had higher BWGAZ; major roads lowered BWGAZ; immigrant enclaves boosted BWGAZ.
Hebert et al. [121]	Examine risk factors for GDM, breastfeeding patterns, and early cessation across racial/ethnic groups.	Focuses on women who responded to the PRAMS survey and gave birth within a specific timeframe.	Breastfeeding rates varied by race, with AI/AN women less likely to start or continue; education boosted initiation.
Roese et al. [167]	Assess SMM factors and disparities among AI/AN pregnant people and develop targeted interventions.	Inclusion: AI/AN individuals linked to the Northwest Tribal Registry; Exclusion: Missing County data or misclassified records.	AI/AN individuals face 1.6x higher SMM rates than NHW, driven by inequities, low PNC, and community factors.
Mancuso et al. [163]	Examine the impact of prenatal mood and anxiety disorders on immune markers and maternal-infant health.	Inclusion: Pregnant women 18+, singleton pregnancy < 16 weeks; Exclusion: HIV+, smokers, substance abusers, multiples.	Mood disorders disrupt inflammation and raise obstetric risks linked to SES disparities.
Barrozo et al. [138]	Map placental transcription and study SARS-CoV-2 dynamics, immune	Inclusion: Pregnant women delivering Aug-Oct 2021 with SARS-CoV-2 tests;	SARS-CoV-2 was found in placentae, linked to inflammation, M2

Table 3 (*continued*)

Articles	Focus	Inclusion/ Exclusion Criteria	Outcomes
	responses, and fetal sex differences.	Exclusion: Those vaccinated during pregnancy.	macrophage loss, and isolated replication sites.
Mead et al. [141]	Evaluate maternal vitamin D levels on placental development, weight, pathology, and outcome disparities.	Inclusion: Women 18–45, singleton pregnancy; Exclusion: Thyroid disease, multiples, sickle cell, sarcoidosis.	Vitamin D and high BMI increased placental weight; Black mothers had smaller placentas.
Fuller et al. [142]	Assess racial disparities in obstetrical outcomes and the impact of universal healthcare.	Focus on deliveries and severe maternal morbidity in the NPIC database.	Racial disparities show higher cesarean and morbidity rates for Black and Asian Pacific Islander groups.
Rose et al. [159]	Investigate prenatal stressors' impact on oxidative stress, race differences, and interventions for birth outcomes.	Inclusion: Women 18+, pregnant < 30 weeks, fluent in English/Spanish; Exclusion: HIV/ hepatitis B/C or fetal demise.	Psychosocial stressors linked to oxidative stress in African American women, with lower gestational age at birth.
Quintero et al. [148]	Estimate breastfeeding information sources by race/ethnicity and their impact on breastfeeding.	Inclusion: Women living with their infant; Exclusion: Missing race, under 18, or missing breastfeeding data.	Breastfeeding rates vary by race, with support boosting durations; structural factors drive disparities.
Chandler-Laney et al. [150]	Evaluate gestational weight gain and perinatal outcomes for Black and White women with OW/OB at prenatal care entry.	Inclusion: Black/ White, BMI 25–45, singleton, 14 weeks; Exclusion: Anomaly, illness, preterm birth, diabetes, substance use.	No racial differences in GWG and perinatal outcomes, indicating reduced disparities at higher BMIs.
Mallampati et al. [151]	Evaluate race/ ethnicity and severe maternal morbidity in two North Carolina healthcare systems.	Analysis included non-Hispanic Black, Hispanic, and non-Hispanic White pregnant individuals.	Non-Hispanic Black and Hispanic individuals have higher severe maternal morbidity rates.
Shara et al. [157]	Evaluate HOPE-CAT's ability to detect maternal cardiovascular conditions and early pregnancy complications.	Inclusion: Women 18–40 with multiple pregnancy visits; Exclusion: Limited EHR data or first record as birth.	HOPE-CAT detected risks earlier, mainly in Black patients, reducing disparities.
Song et al. [152]	Examine pregnancy lipidomic profiles, their link to neonatal measures, and lipid class changes affecting growth.	Inclusion: Pregnant women 18–40, singleton, 8–13 weeks' gestation; Exclusion: Pre-existing conditions or multiple pregnancies.	Triglycerides in early pregnancy increase birth weight, while late pregnancy lipids reduce neonatal size.
Weber et al. [153]	Examine anemia prevalence in pregnancy at Sanford Health, racial/ethnic disparities, and EMR documentation gaps.	Inclusion: Pregnant women 18+, at least one visit to Sanford Health (2018–2020); Exclusion: Incomplete or missing data.	Anemia rates were higher in AI/AN (58.3 %) and Black (55.1 %) patients, with 81.1 % of cases missed, showing racial disparities.
Suplee et al. [155]	Evaluate an online POST-BIRTH Warning Signs project to improve nurses' knowledge and discharge education.	–	Nurses' knowledge rose by 11 %, with improved practice and widespread protocol implementation.

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Table 3 (*continued*)

Articles	Focus	Inclusion/ Exclusion Criteria	Outcomes
Poliektov et al. [156]	Compare maternal serum cytokine profiles by socioeconomic status and assess their link to adverse pregnancy outcomes.	Participants were patients who delivered at specified hospitals, likely requiring pregnancy and prenatal care at these locations.	Low SES is linked to increased inflammation, impacting pregnancy outcomes.
Disparities in Behavioral and Lifestyle Factors			
Brown et al. [52]	Studying ethnic identity, and lifestyle in high-risk populations for chronic diseases, incl. type 2 diabetes	Pregnant women with GDM, aged 18+, from all 44 KPNC facilities, undisclosed exclusions	Stronger ethnic identity is linked to healthier lifestyles, especially in Filipina and Latina women
Oh et al. [102]	Racial/ethnic differences in FAS risk, constant ethanol intake's effect on FAS development	Pregnant women with alcohol consumption; exclude those without verified exposure	Explores racial disparities in prenatal alcohol use and FAS risks for AI/AN, increasing with intake
Specker et al. [60]	Study on tobacco and alcohol use in 2014 South Dakota pregnant mothers	Mothers who had a live birth in South Dakota in 2014, without specific exclusion criteria provided	There were no racial differences in pregnancy smoking or drinking; American Indians had lower rates
Gardener et al. [98]	Examines phthalate/ BPA in pregnant women, diet, disparities by race/ education, and diet's role	Included third-trimester pregnant women without exclusions	Phthalate links to diet, group disparities in levels, mixed results on organic food's impact
Varde et al. [103]	Investigate glyphosate exposure's link to preterm birth and assess potential Black/white disparities	Healthy adult women with uncomplicated pregnancies	Glyphosate might relate to preterm births, potentially affecting Black women more
Theiler et al. [86]	Analyzing COVID-19 vaccine in pregnant patients, comparing outcomes between vaccinated and unvaccinated.	Patients 16–55 delivering at Mayo Clinic Health System, excluding research opt-outs	Vaccinated pregnant women saw fewer infections and no adverse outcomes despite uptake disparities
Barrett et al. [97]	Examines phthalate metabolites and pCRH levels in pregnancy, individually and combined, at two stages	Include: Healthy singleton pregnancies, weeks 16–29, ages 16–40; Exclude: Major medical issues	Phthalate mixtures affect mid-pregnancy pCRH, with exceptions in diabetes and hypertension
Razzaghi et al. [91]	Examining COVID-19 vaccination in women, including pregnancy, attitudes, regions, and healthcare	Women aged 18–49 years, but specific exclusion criteria were not detailed	Pregnant Black women vaccinated less rose after CDC advice; provider guidance drives disparities
Lugo-Candelas et al. [123]	Examine prenatal sleep's impact on offspring ADHD, sleep problems, and neurodevelopment, considering trimester and sex effects.	Inclusion: Mother-child dyads with prenatal sleep and ADHD/sleep data; Exclusion: Missing covariates or incomplete assessments.	Poor second-trimester sleep links to higher ADHD and sleep issues in offspring, highlighting a key time for maternal sleep interventions.
Bodnar et al. [131]	Examine racial discrimination's impact on diet and adherence to dietary guidelines.	Inclusion: Singleton pregnancy, 6–13 weeks gestation, no previous pregnancy ≥ 20 weeks; exclusion not specified.	Racial discrimination did not significantly affect diet quality across groups.
Sanders et al. [132]	Investigate racial inequities in postpartum weight retention, self-weighing behavior, and related factors.	Included: Self-weighing Black or non-Hispanic White participants; Excluded: Missing data or other races.	Self-weighing showed racial income disparities, higher during pregnancy, and dropped with postpartum stress.
Lee et al. [140]	Explore ovarian responsiveness, live birth rates, and IVF disparities by race/ ethnicity.	Inclusion: Patients 18–45 with gonadotropin IVF and race; Exclusion: Missing race or donor oocytes/ carrier.	Black and Hispanic patients had higher OSI but lower LBR and higher miscarriage rates.
Amorim et al. [145]	Investigate the effect of income support on breastfeeding behaviors among urban Alaskan mothers.	Postpartum women in urban Alaska were included, excluding those with missing breastfeeding or race data.	Cash payouts improve breastfeeding, with disparities across socioeconomic and racial groups.
Disparities in Mental Health			
Ojukwu et al. [75]	Predicting peripartum depressive symptoms in Black women with HIV, considering social factors	WLWH aged 18 + in prenatal care at PRIM and WISH, excluding severe health issues	PDS predictors: low income, intimate partner violence, and childcare burden
Saadat et al. [96]	Maternal inflammation, depression in pregnant African American women, risk factors, and health outcomes	Included: African American women 18–45, single pregnancies, English speakers.	Depression, IPV, and smoking in pregnant African American women impact inflammation
Boyer et al. [114]	Identify stressors in pregnant women and evaluate their impact on pregnancy cardiovascular health	Included nulliparous women 13+, singleton pregnancies; excluded missing data	Psychosocial stress is linked to higher cardiovascular risk, driven by obesity, inactivity, and smoking
Goyal et al. [25]	Postpartum depression rates in Asian subgroups vs. White, with demographics	Postpartum depression in high-SES mothers of Black, White, or Hispanic ethnicity	Asian American moms had lower PPD rates than White moms, with age and cesarean sections as risks
Dolbier et al. [42]	Study on postpartum depression in African American and white women, considering socioeconomic factors	Women 18–40, African American or white, with births ≥ 20 weeks, excluding criteria	Race no longer predicted PPD when adjusting for SES and marital status
Park et al. [27]	Study on bias reduction in ML for postpartum depression predictions	Women 12–55, live births, 80 %+ enrollment, mental health services included	Study found racial PPD disparities and varying effectiveness in addressing fairness metrics

Various studies have adopted additional methods to improve model evaluation. For example, Weber et al. [61] correlated and grouped features in their modeling approach, while Sarullo et al. [32] utilized 10-fold cross-validation, using one-fold as the validation set and the others for training, while emphasizing the R^2 value, sensitivity, and specificity. Moreover, Ashrap et al. and Zhang et al. [76,85] used descriptive models to explore psychological and environmental factors, highlighting the

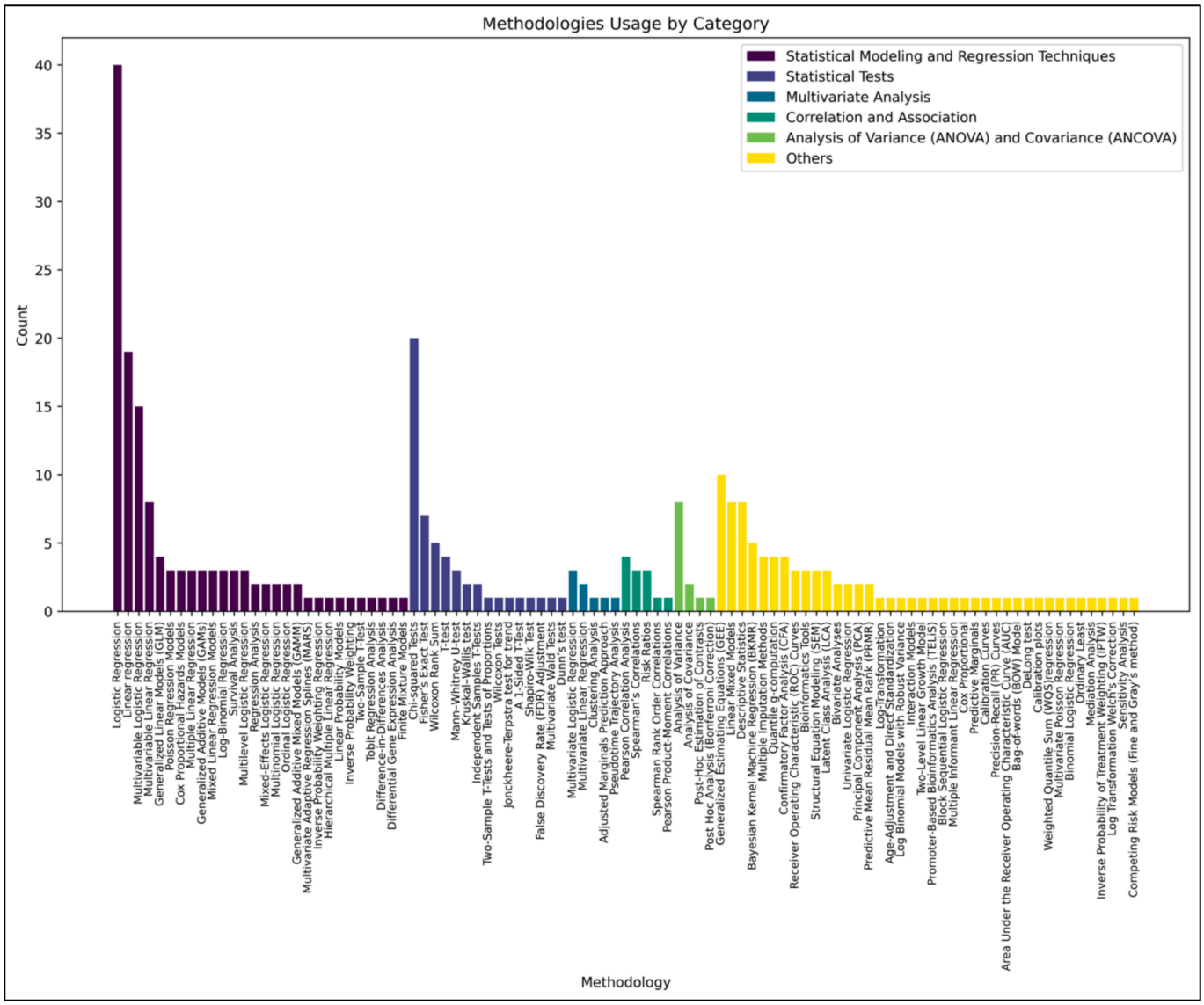

Fig. 3. Distribution of Statistical Methodologies in Maternal Health Disparities Research.

need for tailored approaches in maternal health research. Tables 4 and 5 outline approaches to enhance ML capabilities and summarize algorithms, performance measures, and best models, respectively.

4. Discussion

This systematic review analyzed 147 studies on maternal health disparities, revealing intricate interactions between sociodemographic factors, healthcare methodologies, and health outcomes. It highlights the need to address methodological constraints, data quality issues, and resource allocation in current research. These findings emphasize the importance of considering these elements for their contributions to existing research and guiding future studies. Consequently, the reviewed publications substantially contribute in several key areas to the existing literature in several key areas, highlighting insights and advancing our understanding of crucial health-related concerns.

A substantial amount of research, 59 studies, including Unal et al. [22], Wright et al. [25], Giurgescu et al. [23], Choi et al. [37], and Zhou et al. [70], provide ground-breaking insights on ethnic and racial disparities in health. These studies offer insight into the complex ways in which race and ethnicity intersect with a variety of health outcomes, such as maternal and child health, mental health diagnoses, and environmental pollutants. They underscore the need for culturally sensitive and inclusive health practices and policies. For example, Zhou et al. [70] found that the live birth rates for Black recipients of oocyte donation are significantly lower than those for White recipients. This highlights disparities in reproductive health outcomes and underscores the need for targeted interventions to address these differences. This corpus of work substantially contributes to our understanding of healthcare disparities and outcomes, paving the path for more equitable health solutions.

Building on these findings, studies such as Dolbier et al. [42], Brown et al. [52], Miller et al. [57], Hernandez-Castro et al. [165], and Roese et al. [167], have expanded our understanding of health issues such as preterm birth, gestational diabetes, and the influence of socioeconomic health factors. These studies give more in-depth insights into the complex interplay of numerous factors ranging from racial and socioeconomic disparities, psychological states, environmental settings, and behavioral patterns, with a focus on how these elements collectively influence maternal health outcomes, underlining the need to consider a wide variety of effects in healthcare research. The results of these studies are crucial in guiding more effective, targeted medical interventions and improving overall health outcomes.

In addition, a notable group of studies, including Park et al. [27],

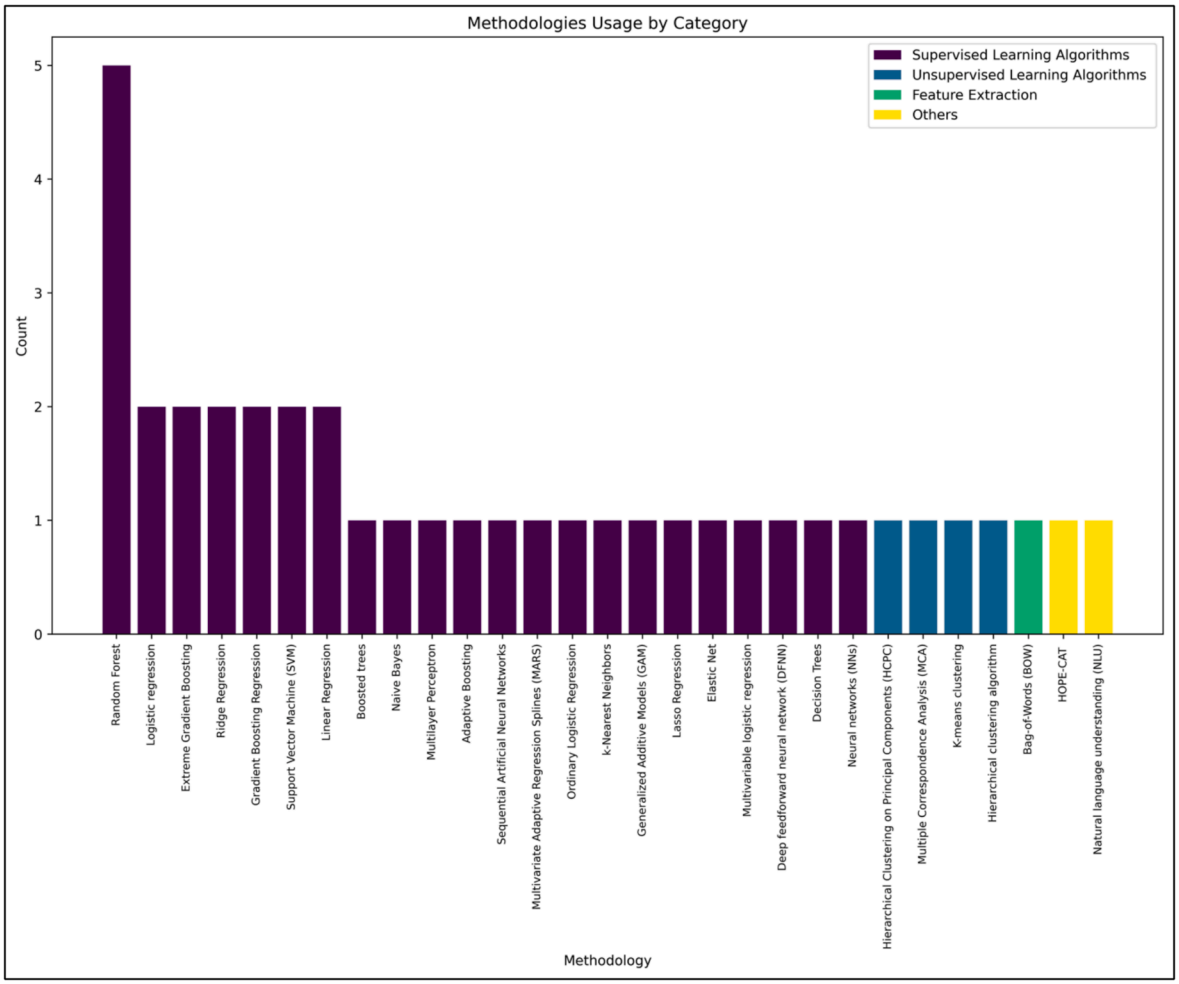

Fig. 4. Count of ML algorithms utilized across the reviewed studies.

Weber et al. [61], Clapp et al. [94], and Barcelona et al. [161], introduce innovative methodological methods or novel applications of existing data. These publications show how novel analytical techniques, data processing methodologies, and innovative data applications can lead to substantial advances in health research. By pushing the boundaries of traditional research approaches, these investigations open new opportunities for addressing complicated health challenges and improving our understanding of many health-related phenomena. For instance, in a study conducted by Park et al. [27], advanced bias mitigation techniques were utilized in ML models to predict postpartum depression more equitably across different racial groups. The researchers significantly enhanced fairness metrics by implementing strategies like reweighing. This demonstrated how innovative data processing methods can improve the accuracy of health predictions and promote fairness. This approach tackles complex health challenges and enhances our understanding of health disparities, paving the way for targeted and effective interventions.

Moreover, our review extends the insights from earlier systematic studies like those by Humphries et al. [10] and Ashman et al. [11], by focusing on personalized and culturally adapted interventions. Utilizing machine learning, it aligns with the targeted approaches of Vats et al. [1] and Bizuayehu et al. [13] to effectively address and reduce disparities related to socioeconomic and geographic factors.

However, despite these advances, the selected studies have led to exploring three primary categories of limitations, which are critical for understanding the scope and relevance of their findings. Approximately 87 studies, including those by Hardeman et al. [79], Zhang et al. [85], and Roese et al. [167], Parikh et al. [158], have substantial methodological and design limitations. These include cross-sectional studies, which limit causal inference, observational and retrospective designs, which are susceptible to biases, and issues with generalizability due to specific subpopulations or single-center designs. These findings underscore the need for more robust, longitudinal study methods to give more accurate causal insights while reducing the risk of biases common in observational and retrospective studies.

Moreover, the problem of assuring data quality and dependability is a common issue in many studies (e.g., Christian et al. [39], Zamora-Kapoor et al. [53], Rose et al. [159], and Shara et al. [157]. The need for more rigorous data collection and validation methodologies is underscored by the reliance on self-reported data, potential misclassification, and the use of administrative or billing data, which may require complete accuracy or comprehensiveness. Improving data quality is critical to the credibility of research findings and may substantially impact their interpretation and use in real-world scenarios.

Furthermore, another common limitation noted in studies such as Unal et al. [22], Wegienka et al. [40], Giurgescu et al. [23], Gerson et al. [68], and Mancuso et al. [163] is concern about sample size and the generalizability of the findings. Many studies, including 20, have limited sample sizes or samples that differ from larger groups, thus limiting the generalizability of results. This suggests that future research should focus on raising sample sizes and ensuring that study populations are varied and representative, improving the applicability and relevance of research findings across different settings and groups.

Despite these challenges, the recommendations from the reviewed studies offer clear pathways for future research. Many studies, including Kitsantas et al. [37], Unal et al. [22], Goyal et al. [25], and Dunlop et al.

Table 4
Overview of Feature Selection and Model Evaluation Techniques in Maternal Health Machine Learning Studies.

Publication	Feature Selection	Cross-Validation Strategy	Hyper-parameter Tuning	Data Splitting	Imbalanced Data Handling	Model Interpretability	Results Validity
Menon et al. [44]	MARS itself performs feature selection.	10-fold cross-validation was used	–	70 % for training and 30 % for testing	–	MARS models are interpretable due to their basis function structure.	Using ROC AUC values
Weber et al. [61]	Features were grouped and highly correlated were collapsed	5-fold cross-validation was used	–	–	–	–	Using assessed through cross-validated AUCs
Koning and Ehrenthal [64]	Using Multiple Correspondence Analysis (MCA)	The HCPC was replicated on random subsamples	–	No splitting (Descriptive Model)	–	Clustering and MCA boost interpretability by grouping based on real-world factors	Supported by the replication of clustering
Park et al. [27]	Based on features relevance	–	–	Split into train, validation, and test sets in a ratio of 5:3:2	Reweighing was used	Logistic regression through its coefficients	Fairness Metrics and Balanced Accuracy were used
Ashrap et al. [76]	Based on psychosocial features	–	Number of clusters (k)	No splitting (Descriptive Model)	–	–	–
Zhang et al. [85]	Based on features relevance	–	–	–	–	–	–
Piekos et al. [87]	Based on features relevance	–	–	80 % for training and 20 % for testing	–	Gini importance and Shapley additive explanations (SHAP) were utilized	–
Clapp et al. [94]	Including only features that occurred within frequency thresholds (≥5% and ≤ 80 %)	Data from hospital B was used for external validation.	LASSO	75 % for training and 25 % for testing	–	–	Using AUC values
Bartal et al. [119]	Sentence-Transformers NLP model was used	10-fold cross-validation was used	Using cycles, batch size, and learning rate variations	–	Using Downsampling	Using Word Category Analysis (LIWC)	Using cross-validation and balanced datasets
Sarullo et al. [32]	–	10-fold cross-validation was used	–	–	–	Holdout Method and Shapley Additive were used	Supported by the significant difference in R^2
Oyebode et al. [107]	–	Time-series and Spatial cross-validation	–	–	–	–	–
Cersonsky et al. [109]	Based on the literature review and Exploratory Factor analysis (EFA)	100 k-fold cross-validation was used	Tuning matrices unique to each model	Training and testing (3:1 Ratio)	SMOTE was used	Variable importance was reported for each model	Using AUROC
Mane et al. [129]	–	–	–	–	–	–	–
Piekos et al. [139]	Gini-based importance and Shapley Additive Explanations (SHAP) values.	5-Fold Cross-Validation	Number of estimators (trees) Maximum depth of trees Minimum samples per leaf	–	Propensity score matching was used	Gini-Based Importance and SHAP Values	Cross-validation with AUC scores.
Ren et al. [144]	–	5-Fold Cross-Validation	–	–	Random undersampling, random oversampling, synthetic minority oversampling technique (SMOTE), and weight rebalancing	–	Extensive experimentation using cross-validation and compared performance across multiple rebalancing
Nguyen et al. [21]	–	5-Fold Cross-Validation	–	–	–	–	Accuracy and F1-score
Barcelona et al. [161]	Information Gain was applied	–	–	70 % for training and 30 % for testing	–	Decision Trees offer clarity; SVM and Random Forests handle complexity.	–
Shara et al. [157]	–	–	–	–	–	HOPE-CAT interprets risk by analyzing encounter-by-encounter data trends for each participant.	Results were validated against clinical diagnoses recorded in the EHR.

Table 5
Summary of Machine Learning Algorithms, Performance Measures, and Best Models in Maternal Health Studies.

Publication	Algorithms Name	Performance Measures	Most Effective Model
Predicting outcomes such as spontaneous preterm birth			
Cersonsky et al. [109]	Boosted trees and Random forests	Boosted trees: AUROC 0.867, Sensitivity 86.9 %, Specificity 86.4 %, Positive Predictive Value 89.5 %, and Negative Predictive Value 83.2 % Random forests: AUROC 0.850, Sensitivity 92.2 %, Specificity 77.9 %, Positive, Predictive Value 84.7 %, and Negative Predictive Value 83.2 %	Boosted trees
Ren et al. [144]	Logistic regression, Naive Bayes, Random Forest, Extreme Gradient Boosting, Adaptive Boosting, Multilayer Perceptron, and Sequential Artificial Neural Networks	Logistic Regression (Avg Recall = 0.51) Naive Bayes (Avg Recall = 0.38) Random Forest (Avg Recall = 0.29) Extreme Gradient Boosting (Avg Recall = 0.46) Adaptive Boosting (Avg Recall = 0.39) Multilayer Perceptron (Avg Recall = 0.47) Sequential ANN (Avg Recall = 0.50)	Extreme Gradient Boosting
Shara et al. [157]	HOPE-CAT (Healthy Outcomes for all Pregnancy Experiences – Cardiovascular Risk Assessment Technology)	Delta time before intervention	–
Early spontaneous preterm birth in specific racial groups			
Menon et al. [44]	Multivariate Adaptive Regression Splines (MARS)	Using ROC AUC values	–
Weber et al. [61]	Ordinary Logistic Regression, Random Forest, k-Nearest Neighbors, Generalized Additive Models (GAM), Lasso Regression, Ridge Regression, and Elastic Net.	Sensitivity = 0.61 Specificity = 0.64 Predictive Value Positive (PVP) = 0.017 Predictive Value Negative (PVN) = 0.994 For all models	all models achieved similar performance with low predictive accuracy.
The impact of maternal life events and stress on birth outcomes			
Koning and Ehrenthal [64]	Hierarchical Clustering on Principal Components (HCPC), and Multiple Correspondence Analysis (MCA)	Eigenvalues Variance Explained	–
Ashrap et al. [76]	K-means clustering	–	–
Nguyen et al. [21]	Support Vector Machine (SVM)	Accuracy = 91 % F1 Score = 84 %	–
The effects of maternal SARS-CoV-2 infection			
Piekos et al. [87]	Linear Regression, Ridge Regression, Gradient Boosting Regression, and Random Forest	Coefficient of Determination (R^2 value), Root Mean Square Deviation, and Accuracy	Random forest
Piekos et al. [139]	Gradient Boosting Regression	Vaccination Status Model: AUC = 0.81 (Full model). AUC = 0.79 (Limited model with top 5 features). Booster Status Model: AUC = 0.73 (Full model). AUC = 0.72 (Limited model with top 5 features).	–
Identifying Urban Environmental Factors Influencing Prenatal Care			
Zhang et al. [85]	Hierarchical clustering algorithm	Silhouette values, Chi-square and ANOVA	–
Oyebode et al. [107]	Multivariable logistic regression	–	–
Reducing bias in predicting postpartum depression			
Park et al. [27]	Logistic Regression, Random Forest, and Extreme Gradient Boosting	The balanced accuracy was 0.73 for Postpartum Depression Prediction and 0.78 for Mental Health Service Utilization across all models.	All models achieved similar performance
Clapp et al. [94]	Bag-of-Words (BOW)	AUC = 0.72 (internal validation) AUC = 0.76 (external validation)	–
Bartal et al. [119]	Deep feedforward neural network (DFNN)	AUC = 0.75, F1-score = 0.76, Sensitivity = 0.8, and Specificity = 0.70	–
Mane et al. [129]	Natural language understanding (NLU)	Accuracy = 98.5 %	–
Barcelona et al. [161]	Decision Trees Random Forest Support Vector Machines	For Marginalizing Language: Decision Trees: Precision = 0.72, Recall = 0.73, F-Score = 0.73. Random Forest: Precision = 0.72, Recall = 0.70, F-Score = 0.71. Support Vector Machines: Precision = 0.63, Recall = 0.65, F-Score = 0.64. For Power/ Privilege Language: Decision Trees: Precision = 0.86, Recall = 0.86, F-Score = 0.86. Random Forest: Precision = 0.87, Recall = 0.77, F-Score = 0.81. Support Vector Machines: Precision = 0.88, Recall = 0.95, F-Score = 0.91.	Decision Trees excelled at identifying marginalizing language, while Support Vector Machines were most effective for detecting power/ privilege language.
Examining how socioeconomic factors and race affect fetal development			

(*continued on next page*)

Table 5 (*continued*)

Publication	Algorithms Name	Performance Measures	Most Effective Model
Sarullo et al. [32]	Linear regression (LR) Neural networks (NNs)	$R^2 = 0.145$ $R^2 = 0.172$	Neural networks (NNs)

[38], underline the need for more studies to confirm and extend their results. This involves confirming current findings in broader or more diverse groups, exploring underlying mechanisms, and broadening research to include new areas or populations. This ongoing exploration is critical for solidifying an understanding of these complex challenges and ensuring that research findings are robust and relevant in various contexts.

Additionally, studies such as Kitsantas et al. [37], Goyal et al. [25], Wegienka et al. [40], and Roese et al. [167], call for the development of targeted interventions, clinical strategies, and public health approaches. These proposals highlight the necessity of translating research results into practical applications that directly influence health outcomes. Future research can extend beyond understanding health phenomena to actively improve healthcare practices and policy by concentrating on intervention and strategy creation. For example, Kitsantas et al. [37] found that providing personalized breastfeeding support can significantly improve breastfeeding results for Hispanic women with low socioeconomic status. These women tend to have higher rates of starting breastfeeding but encounter difficulties in sustaining it. This discovery underscores the importance of interventions tailored to their cultural and socioeconomic context.

Several studies, including Dunlop et al. [38], Christian et al. [39], Menon et al. [44], and Trasande et al. [162], emphasize the need for methodological changes and the adoption of new analytical methodologies. These studies highlight the benefits of longitudinal study designs, more diversified populations, and sophisticated statistical techniques. Such methodological advances are critical for overcoming the constraints of present research methodologies and gaining deeper, more accurate insights into health-related issues.

Finally, there is a clear call from studies like Goyal et al. [25], Dolbier et al. [42], Zahran et al. [24], and Nguyen et al. [21] to expand research efforts to cover underrepresented or understudied populations and areas. This expansion is necessary to ensure that study outcomes are inclusive and reflect diverse populations. Addressing these research gaps can lead to a more comprehensive understanding of health concerns and the ability to customize health care and treatments to the needs of all population segments. To gain insight into the specific socioeconomic and racial influences that contribute to mental health discrepancies in these areas, Dolbier et al. [42] highlighted the importance of studying postpartum depressive symptoms in rural African American women. This group is often underrepresented in research.

Healthcare providers can use ML insights, as highlighted in this review, to design tailored interventions for high-risk maternal populations. By analyzing demographic, socioeconomic, and clinical data, ML models identify at-risk groups and generate personalized risk scores, enabling targeted resource allocation, early interventions, and preemptive care. These strategies translate into actionable solutions to reduce disparities and improve maternal health outcomes.

5. Conclusion

This systematic review demonstrates the potential for transforming maternal healthcare through machine learning (ML) and statistical methods. Insights from complex dataset analyses have highlighted disparities across racial, socioeconomic, and geographic lines. While there have been notable advancements, optimizing these methods is still needed. Recommendations include enhancing dataset diversity, improving methodological precision, and adopting interdisciplinary approaches. In particular, integrating social determinants of health, such as housing, education, and community conditions, into predictive models could capture a broader range of risk factors and promote equitable healthcare outcomes. Further efforts should focus on using ML and statistical advancements to improve maternal health outcomes and integrate these tools into policymaking and practice, adhering to ethical standards and emphasizing health equity.

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Hala Al Sliti: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Ashaar Ismail Rasheed:** Writing – review & editing, Validation, Data curation. **Saumya Tripathi:** Writing – review & editing. **Stephanie Tulk Jesso:** Writing – review & editing. **Sreenath Chalil Madathil:** Writing – review & editing, Supervision.

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During the preparation of this work, the author(s) used [OpenAI/ChatGpt] in order to [improve language and readability]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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